



Monitoring report – Lake Taney

Sampling and sensing aquatic ecosystems, from 1D to 3D

Spring semester - 2026



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Abstract

This report presents the first annual follow-up of Switzerland's first student-led lake monitoring program, established at Lake Taney (Valais) in 2024. Building on a vertical sensor mooring deployment at the lake's deepest point (~35 m), we analysed one full annual cycle (June 2025-June 2026) of temperature, dissolved oxygen, pressure, and meteorological data, complemented by CTD and bathymetric transects conducted in June 2026. Lake Taney behaves as a dimictic lake, with clear summer stratification, winter ice cover, and two mixing events per year. Bottom-water oxygen, however, remained below the Swiss legal ecological threshold (4 mg/L) for 83.7% of the monitored year, including a 147-day near-anoxic episode in summer-autumn 2025, and spring mixing in 2026 was too brief and incomplete to reoxygenate the deep layers. An exponential decay model applied to the under-ice period ($k = 0.039 \text{ day}^{-1}$) places Taney among the faster-depleting alpine lakes reported in the literature. These results point to a lake whose surface water appears oligotrophic, but whose deep waters show meso- to eutrophic characteristics, a condition likely to worsen under climate warming through longer stratification and shorter, less effective ice-driven mixing. We identify key methodological limitations from this year's campaign, notably single-depth oxygen sensing, gaps in the pressure record, and implement mooring modifications and monitoring recommendations to strengthen the interpretability of future campaigns.

1. Introduction

1.1 Context

Lakes are particularly sensitive indicators of climate variability and environmental change, as their thermal and biogeochemical dynamics respond rapidly to atmospheric forcing (Woolway et al., 2020; Williamson et al., 2009). To characterize the physical and biogeochemical dynamics of Lake Taney, a vertical sensor array was deployed along the deepest axis of the lake, estimated at 35 m depth (Russo et al., 2025), at coordinates 554061 E / 132893 N (Swiss coordinate system LV95). The array comprises temperature, pressure, and dissolved oxygen sensors distributed along the water column to obtain continuous vertical profiles.

This present report constitutes the first annual follow-up of this long-term observational program, building on data collected between April and December 2024 and between May and June 2025.

1.2 Study area characteristics

Lake Taney is a natural mountain lake situated at an altitude of 1,409 m a.s.l. within the perimeter of the Federal Inventory of Landscapes, Sites and Natural Monuments of National Importance (IFP No. 1702), in the municipality of Vouvry, Chablais region, canton of Valais, Switzerland (Figure 1). Nestled between the summits of the Grammont (2,171 m a.s.l.) and Les Jumelles (2,215 m a.s.l.), the lake occupies a glacially eroded basin and drains into the hydrological catchment of the Rhône River. It covers a surface area of 0.17 km² and reaches a maximum summer depth of 35 m (Russo et al., 2025).

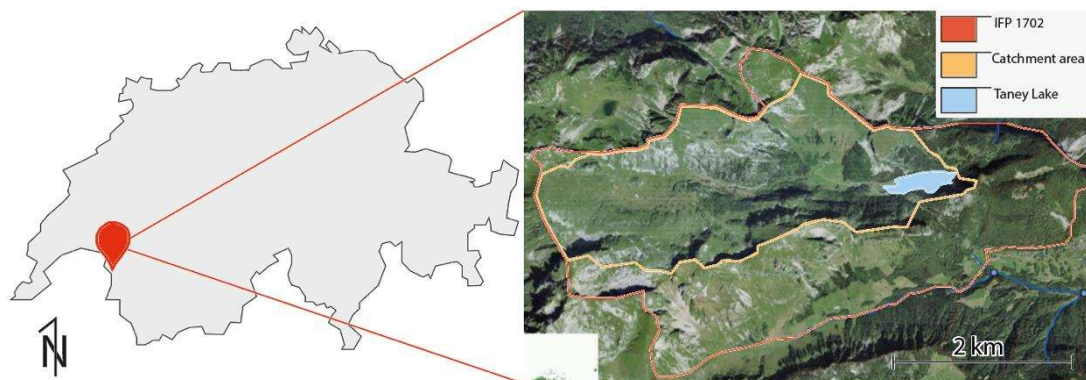


Figure 1: Location of Lake Taney, its catchment area (yellow) and the IFP-listed perimeter (red). Base map: Swisstopo (2025), SwissBoundaries3D. Catchment boundaries: Swisstopo (2025). IFP perimeter: map.geo.admin.ch, Bundesinventar Nr. 1702. Imagery: Swisstopo (2025). Author: Quentin Noël.

1.3 Hydrological context

The hydrological regime of Lake Taney is notably complex. The lake receives surface inflow from a single tributary entering from the western shore, while no natural surface outflow exists. Subsurface drainage occurs through a karstic network connecting the lake to the adjacent Tové valley, and an overflow on the eastern shore activates at high water levels. A bottom water intake regulates the lake level for hydroelectric purposes and is shut down each spring during amphibian spawning season. As a result of these combined natural and artificial controls, the lake undergoes managed water level fluctuations (OFEV, 2022).

The lake measures 940 m in length and 270 m in width, with a surface area of 0.17 km² and a total water storage capacity of 2.60×10^6 m³ (geo.admin.ch, 2022; OFEV, 2022). The catchment spans elevations from 1,398 m to 2,297 m a.s.l. (geo.admin.ch, 2022), with a mean elevation of 1,823.5 m a.s.l., entirely Swiss territory (100%).

1.4 Conservation history and current anthropogenic pressures

Lake Taney has been a protected natural reserve since 1966. Through a formal agreement between the *Ligue Valaisanne pour la Protection de la Nature* (LVPN), *Patrimoine Suisse*, the *Swiss Alpine Club* (CAS), and the *bourgeoisie of Vouvry*, the Taney-Grammont region was placed under formal protection, shielding it from inappropriate development, construction, and motorised traffic (Gabbud, 2021). This partnership, originally valid for fifty years, was renewed for a further then-year period by the same parties, now including *Pro Natura* and the municipality of Vouvry (Pro Natura, n.d.; Vouvry, 1967).

The site is inscribed in the *Federal Inventory of Landscapes, Sites and Natural Monuments of National Importance* (IFP) under the designation Lac de Taney (Canton du Valais, 2018) and is listed in the Federal Inventory of Amphibian Breeding Sites of National Importance, supporting several specialized species including the common toad (*Bufo bufo*) (OFEV, 2025).

Pressures on the lake operate across two interconnected dimensions. On the anthropogenic side, expansion of tourism risks increasing nutrient inputs through surface runoff, causing shoreline disturbance, and gradually degrading water quality, potentially accelerating eutrophication. Vehicle access is already restricted to authorized users only, and camping has been prohibited since 2016, yet these measures may prove insufficient as visitor numbers grow. On the climate side, rising air temperatures are expected to intensify and prolong summer thermal stratification, reduce deep-water dissolved oxygen availability, and potentially shift the lake away from its current dimictic regime toward incomplete seasonal mixing (Pilla et al., 2020; Woolway et al., 2020). Insufficient mixing limits oxygen renewal in the hypolimnion, promoting anoxia and phosphorous release from sediments, further accelerating eutrophication in a self-reinforcing cycle. Such a regime shift would have cascading

consequences for biodiversity, including the nationally protected amphibian populations at the site (OFEV, 2025).

1.5 Rationale, objectives and working hypothesis

Systematic monitoring of Lake Taney is essential to track its ecological state over time, detect early signs of degradation, and provide the scientific basis for management decisions aligned with FOEN protection objectives. This report represents the first annual follow-up of Switzerland's first student-led lake monitoring program, established in 2024. As the 2026 team, we are the first group to assess lake conditions using the full mooring dataset, enabling a comparative analysis against the existing record.

1.5.1 Objectives

The 2026 campaign pursues two complementary objectives:

1. **Analyse existing monitoring data.** Retrieve and analyse data to characterise thermal structure, dissolved oxygen, and oxygen saturation, dynamics, with interannual comparison limited by the available historical record (April-December 2024, May-June 2025).
2. **Enhance the monitoring program.** Identify weaknesses in the current approach, implement targeted mooring improvements, and formulate recommendations on sensors, methods, and protocols for future teams.

Both objectives are framed around the following FOEN protection goals: preserving the ecological function of wetland habitats, the natural character of the lake and its aquatic and riparian ecosystems; and the natural dynamics of the watercourse.

1.5.2 Working hypothesis

Based on its altitude, latitude, and 2024-2025 preliminary data, Lake Taney is expected to behave as a dimictic lake, undergoing two full mixing events per year. In summer, thermal stratification develops as solar radiation heats the surface layer, forming a warm epilimnion above a cold, stable hypolimnion. Surface temperatures are expected to reach 15-17°C, while bottom temperatures remain close to 4°C, near the temperature of maximum water density. In winter, ice cover induces inverse stratification, with the coldest water on the surface. Full mixing is expected in spring and autumn as surface and deep-water temperatures converge, and wind-driven circulation.

Dissolved oxygen is expected to follow this seasonal pattern: during both stratified periods (summer and winter), the hypolimnion should progressively be depleted as microbial decomposition outpaces resupply from the surface, with recovery during spring and autumn mixing. The extent to which mixing fully reoxygenates the deep basin each year is one of the key questions this monitoring program aims to track over time.

2. Methodology

2.1 Field site, mooring configuration and transect overview

Lake Taney's monitoring system is centred on a fixed vertical sensor array deployed at the deepest point of the lake (~35 m), at coordinates 554061 E / 132893 N (LV95). The array provides continuous temperature, pressure, and dissolved oxygen measurements throughout the water column.

Two successive mooring configurations have been operated since 2024: the "old mooring" (15 March 2024 - 5 June 2025), and the "new mooring" (6 June 2025 – 3 June 2026, retrieved by the 2026 campaign).

Transects are additionally conducted across the lake to capture spatial variability of the previously cited variables. The first transect (T2) consisted of 15 stations distributed along the transversal axis of the lake, extending from the vicinity of the hydropower station outflow to the cliff on the opposite shore. The second transect (T3) included 14 stations along a different axis (Figure 2). Together, these transects provided complementary cross-sections of the lake and allowed the spatial consistency of thermal stratification and conductivity patterns to be compared with measurements from the fixed mooring.

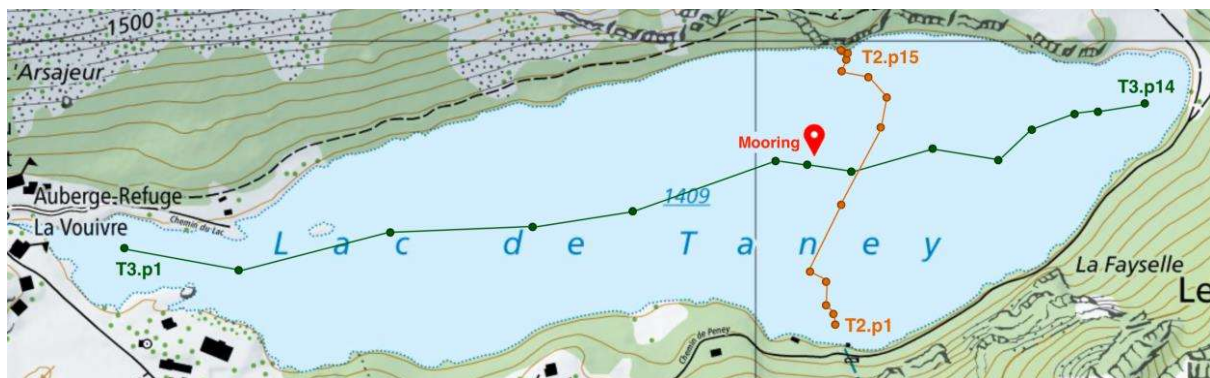


Figure 2: Map of Lake Taney showing the two CTD sampling transects. Orange dots (T2, Transect 2): 15 measurement points. Green dots (T3, Transect 3): 14 measurement points. Each point is the midpoint of a CTD cast used to obtain vertical profiles of temperature, conductivity and depth. Red marker: mooring location.

2.2 Old mooring (March 2024 - June 2025)

The old mooring consisted of a surface buoy connected to a submerged buoy (defined as Level 0) above a bottom weight. Temperature was recorded by ten data loggers mounted along the line, positioned at 1.70, 4.30, 7.13, 9.93, 12.53, 15.20, 17.86, 20.42, 22.97, and 27.92 m below Level 0. Logger depths were initially referenced from the bottom weight and later converted to a top-down scheme for consistency. No pressure or dissolved oxygen sensors were included. See Annex 1 for diagram for mooring 2024 campaign.

2.3 New mooring (June 2025 - June 2026)

The new mooring consisted of two interconnected buoys (Figure 3). Buoy A served as the surface reference, rigged to a 15 m line with a suspended weight; Buoy B acted as a stable submerged reference, rigged to a 20 m line with a bottom. The buoys were connected by a 15 m rope fitted with an auxiliary float to maintain tension. Sensor positions on Buoy A are expressed as depth from the surface; positions on Buoy B, as height above the lake bottom.

On Buoy A, HOBO temperature loggers were installed at 0.5, 1.5, 3.5, 5.5, 7.5, 10.5, and 12.5 m depth, with a pressure sensor at 14 m. On Buoy B, HOBO loggers were mounted at 11.4, 15.4, and 17.5 m above the bottom, with a pressure sensor at 5.4 m and a dissolved oxygen sensor (MiniDOT) at 1.4 m above the bottom. The sensors were referenced in UTC, and local deployment time is accounted for post-processing. At deployment, two Secchi depth measurements were taken (4.70 m and 4.90 m) to characterise light penetration and photic zone.

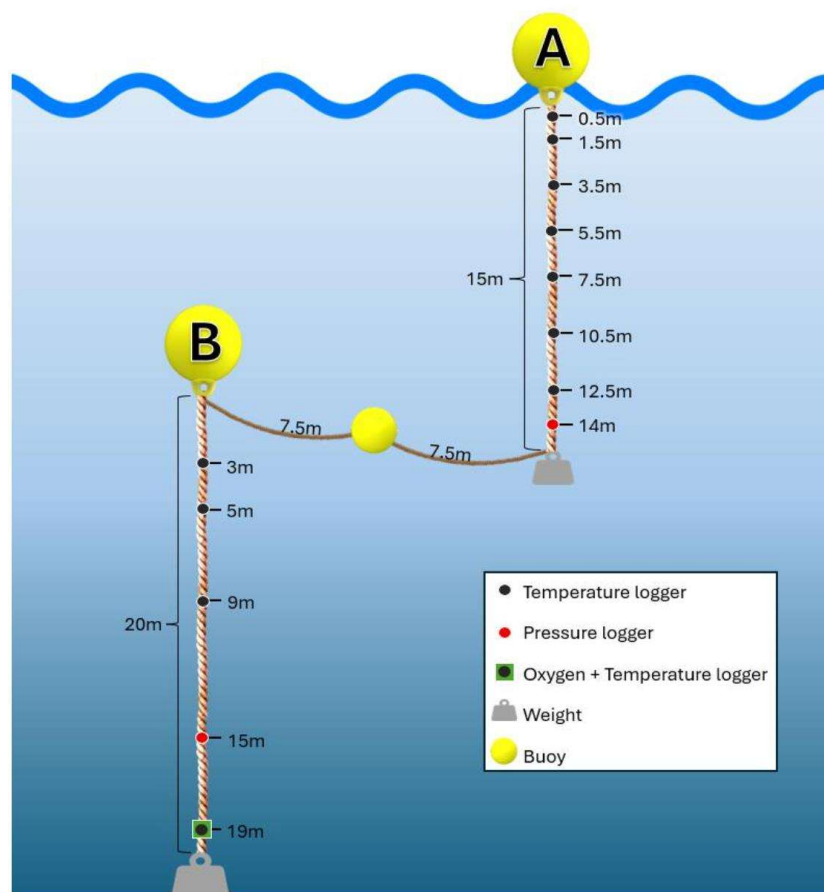


Figure 3: Mooring diagram, 2025 campaign, showing two Buoy A and buoy B lines. Logger IDs encode depth. Temperature logger IDs on buoy A: 21724003_0.5; 21723994_1.5; 21723995_3.5; 21712001_5.5; 21723997_7.5; 21724002_10; 21723999_12.5. Temperature logger IDs on buoy B: B_3, B_5, B_9. Pressure loggers: 20598413_A_14 on buoy A and 20598412_B_15 on buoy B.

2.4 Fieldwork performed (2026 campaign)

2.4.1 Mooring retrieval and redeployment

The 2026 fieldwork focused on the retrieval and servicing of the mooring system deployed in June 2025. Upon retrieval, sensor depths and distances along the mooring line were measured and verified against the labels assigned during the previous deployment. Raw data files were exported directly from each logger.

2.4.2 Transects

Field measurements of CTD and Bathymetry were conducted on the two transects on 4 June 2026 under partly cloudy conditions with intermittent sunny spells. Air temperature was approximately 16.5°C in the afternoon during transect operations. Wind conditions were notable, causing significant lateral drift of the survey vessels throughout both transects, which is reflected in the bathymetric data processing described later.

2.4.2.1 Conductivity Temperature Density (CTD)

CTD measurements were carried out to investigate the spatial variability of temperature, conductivity, oxygen, pressure, and water depth using RBR profilers.

At each station, the CTD probe was attached to a 35-37 m rope and lowered vertically from the boat at an approximate speed of 1 m s⁻¹ while recording continuously. The probe was retrieved once it reached the lake bottom. The midpoint of each cast was used as the representative location for spatial analyses. This procedure provided measurements throughout the entire water column, from the surface to the sediment interface, and enabled the vertical structure of the measured variables to be characterized along both transects.

2.4.2.2 Bathymetry

To complement the CTD measurements, a sonar device was deployed during the T2 transversal transect. The instrument was operated from the vicinity of the hydropower station outflow to the final CTD station located near the cliff face, providing continuous water depth measurements along the transversal axis of the lake and yielding a high-resolution bathymetric profile between successive CTD stations.

The instrument used was a Deeper Smart Sonar Pro+ 2 (model A2AD), floating and towed from the rear-left side of the boat at the surface. The device records water depth (in meters, from surface to lake bottom) at approximately 0.1-second intervals, while GPS position is logged every 10-20 seconds. Surface water temperature is also recorded with the same interval of the water depth. Timestamps have been stored as milliseconds elapsed since 1 January 1970 (Unix epoch).

3. Data analysis

All data processing followed a standardized four-level pipeline, with the corresponding processing Python scripts available in the project GitHub repository: <https://github.com/tdoda/taney-monitoring-masterfieldcamp>

Raw data were first exported directly from the loggers in sensor-specific file formats (Level 0), then cleaned and converted to NetCDF (.nc) format (Level 1). At Level 2, data were corrected and merged for analysis, including metadata attribution, time alignment, and computation of hourly and daily averages. An optional Level 3 includes interpreted or aggregated data used for the final analyses presented in this report. The specific processing steps applied to each dataset, based on this general pipeline, are detailed in the following sections.

All processing levels (Levels 0–3) data are available at:

<https://drive.switch.ch/index.php/s/yyV253N9xIIceSs>

3.1 Weather data analysis

Meteorological data came from two sources. The main dataset came from a local weather station at the Refuge du Grammont, near Lake Taney at 1417 m altitude. This dataset covered 1 October 2024 to 21 May 2026, with one record per day at 12:00 local time. It included air temperature, pressure, wind direction, wind speed, gust speed and precipitation.

The second dataset came from the ICON weather model, at coordinates 46.345857° N and 6.842092° E. It covered 1 June 2025 to 3 June 2026, at hourly resolution, and included 2 m air temperature, wind components, global radiation, relative humidity, sea-level pressure and cloud cover.

The two datasets were not treated as interchangeable. The station gives local observed conditions near the lake, while ICON gives gridded model estimates for the wider area. This distinction matters for variables like radiation, wind and cloud cover, which can vary strongly in mountain terrain.

3.2 Pressure data analysis

Pressure was recorded by two HOBO loggers on the mooring: sensor 20598413 on Chain A at 14 m depth, and sensor 20598412 on Chain B, positioned 5.4 m above the lake bottom. Because lake level changed over time, the actual depth of Sensor B below the water surface was not constant. Both sensors recorded absolute pressure and temperature every 15 minutes.

Raw timestamps and pressure values were matched with deployment metadata, including serial number, nominal position, and deployment and retrieval times. Water-level variation was derived from the pressure difference between Sensor B and Sensor A. Since both sensors were submerged, the atmospheric pressure contribution

cancels out in the difference. The pressure difference was converted into an equivalent water-column height using the hydrostatic relationship ($P = \rho gh$, $\rho = 1000 \text{ kg/m}^3$, $g = 9.81 \text{ m/s}^2$) and expressed relative to the mean pressure difference over the available record. The resulting water-level variation time series was used to correct the depth of moored sensors over time, particularly for Chain B. This correction was only available from 6 June 2025 to 18 January 2026, as the pressure logger stopped recording before the mooring was retrieved on 3 June 2026.

3.3 CTD data analysis

The raw files (.rsk) from the profilers were extracted and matched to metadata file containing the deployment station information: The device used, the people in charge, the lake's name, local time of measurement, coordinates in the Swiss CH1903/LV03 system, and comments.

A quality control was performed on each variable (Time, Pressure, Temperature, Conductivity, Dissolved oxygen) against physically plausible bounds, and values outside these ranges were flagged. The main filter ranges are [0;310] dbar for pressure, [-5; 40] °C for temperature, [0;100] mS/cm for conductivity, and [0;20] mg/L for dissolved oxygen.

For Level 2 processing, flagged values were replaced with NaN, and derived variables were computed: salinity from temperature and conductivity, water density and potential density, depth from the pressure, potential temperature, and oxygen saturation recalculated against a theoretical solubility value.

3.4 Bathymetry data analysis

Bathymetric data from the T2 transect were exported in CSV format. Timestamp conversion from Unix epoch (milliseconds) to local time (UTC+2) was applied as a first processing step.

The analysis pipeline is structured in two stages. The first stage operates on the raw data as recorded directly by the Deeper Pro+ 2, without spatial correction. Inspection of these raw outputs revealed redundancy in the spatial distribution of measurements, likely attributable to lateral drift of the vessel relative to the intended linear transect trajectory. The second stage applies a spatial smoothing procedure, in which GPS positions are projected orthogonally onto a straight reference line defined between the start and end points of the transect. Depth, surface temperature, and position are subsequently averaged within 1-metre bins along this reference axis. This approach mitigates the influence of lateral boat drift induced by lake currents, which would otherwise introduce spatial redundancy and inflate the apparent distribution of measurements. The 15 CTD deployment positions along the T2 transect were

superimposed on the resulting visualizations, providing a spatial reference for each cast and enabling assessment of measurement representativeness along the transect.

A total of 43,568 measurements were recorded along the T2 transect between 11:27 and 12:30 (local time, UTC+2). Water depth ranged from 0.19 m to 35.70 m, with a mean depth of 18.74 ± 11.34 m, consistent with significant bathymetric variability along the transversal axis of the lake. The maximum recorded depth of 35.70 m agrees with the literature value of 35 m (Russo et al., 2025). Surface water temperature remained largely stable throughout the transect, ranging from 16.40 °C to 17.00 °C, with a mean of 16.79 ± 0.16 °C. However, these temperature values should be interpreted with caution, as they are inconsistent with concurrent CTD measurements. This discrepancy is likely attributable to solar heating of the instrument housing: the Deeper Pro+ 2 features a black casing and was exposed to direct sunlight while floating at the air-water interface. This may produce artificially elevated surface temperature readings.

Following spatial smoothing, the straight-line transect length was estimated at 222 m, consistent with the official lake width. Maximum lateral drift relative to the reference axis reached 84.7 m, reflecting the influence of currents on vessel trajectory during the survey. The 43,568 raw measurements were reduced to 178 spatially representative points at 1-metre resolution, with a maximum recorded depth of 34.59 m and a mean surface temperature of 16.85 °C, in agreement with the raw dataset.

3.5 Temperature data analysis

Temperature data from the 2025–2026 period were processed through all four levels of the pipeline presented earlier.

Each thermistor file uses the MM/DD/YYYY HH:MM:SS format in local civil time. The file header states CEST, but this label covers the full deployment, including the winter months under CET. For this reason, timestamps were kept as naive local time, without conversion. At Level 1, measurements outside the deployment window or outside the plausible range of 0 to 30 °C were flagged. The autumn clock change on 26 October 2025 produced six duplicate timestamps per file. Only the first occurrence was kept in each case. The spring clock change on 29 March 2026 created a 70-minute gap, which was left unfilled. At Level 2, flagged values were replaced by NaN. All values after 12:00 on 3 June 2026 were also masked, to remove artefacts from retrieval, since sensors pass through non-nominal depths as the mooring is pulled up. The cutoff time was chosen from a diagnostic plot of temperature against time before the end of deployment.

Level 2 files were linearly interpolated onto a 0.1 m depth grid, at each 10-minute timestep. No extrapolation was applied beyond the sensor range. Chain A depths were fixed at their nominal values. Chain B depths were computed as 31.0 m minus the height above the bottom, corrected for water-level variation where pressure data

allowed. The MiniDOT record was interpolated onto the same time axis and added as the deepest sensor, 14 m below S2.

Chain B depths could only be corrected using pressure data until 18 January 2026. From that date through the ice-cover period, from 16 December 2025 to 10 April 2026, water level was held constant, and Chain B depths were frozen at their last measured value. Outside the ice period, depths were instead estimated by matching each Chain B temperature to the Chain A profile and assigning the depth of the closest match. A constant separation of 2.1 m between S2 and S3 was enforced throughout. This creates a circularity from January 2026 onward: Chain B depths are derived from Chain A temperatures, but those same temperatures are then used to describe the thermal structure at the estimated depths. Because of this, results below 12.53 m carry extra uncertainty, and this region is shown at reduced opacity in all figures.

The water column was classified as stratified when the surface-to-bottom temperature difference exceeded 1°C for at least 48 consecutive hours (Woolway et al., 2014). The water column was classified as ice-covered when persistent inverse stratification was observed. An ice event began when the bottom-minus-surface difference exceeded 0.1°C for at least 48 hours (Pierson et al., 2011) and ended when it fell back below 0°C for the same duration. These asymmetric thresholds were chosen to stop short recrossing from splitting one continuous ice season into several events. Only the longest event was held. Stratification and ice cover were treated as separate, non-overlapping states.

Thermocline depth was defined as the depth of maximum $|dT/dz|$ on the 0.1 m grid. It was computed using central differences, with a one-hour smoothing window applied first, and only during open-water stratified timesteps.

Net atmospheric heat flux was estimated from five components: incoming shortwave radiation, incoming longwave radiation, outgoing longwave radiation, latent heat, and sensible heat. Radiation data came from MeteoSwiss (Aigle station) and from NCEP-DOE Reanalysis 2 cloud-cover data (Kanamitsu et al., 2002). Meteorological forcing came from the local weather station. Outgoing longwave radiation was calculated using the Stefan-Boltzmann law, based on the temperature of the shallowest thermistor. Turbulent fluxes were calculated using bulk aerodynamic formulae, based on wind speed at 10 m. During ice-covered periods, surface heat exchange was assumed to be negligible.

3.6. Dissolved oxygen data analysis

Unlike the other sensors, the MiniDOT stores timestamps directly in UTC, so no time conversion was needed. Where clock drift was measured at retrieval, a fixed offset was added to correct the timestamps. Only oxygen concentration, saturation, and temperature were kept from the raw file; other columns, such as battery voltage, were discarded.

Measurements outside the deployment window, meaning before deployment or after retrieval, were flagged and later replaced by NaN. This made it possible to separate periods when the logger was recording in air. No physical range check was applied to the oxygen values themselves. Metadata such as coordinates, deployment and retrieval times, serial number and depth were added at this stage. Daily mean oxygen concentrations were then computed for the depletion and exceedance analyses below.

An exponential decay model, $DO(t) = DO_{init} \times \exp(-k \cdot t)$, was fitted to two depletion periods, following Perga et al. (2023). The first period was summer stratification, starting at deployment on 6 June 2025. The second was the winter under-ice period, starting at the post-mixing oxygen maximum on 26 December 2025.

Exceedance statistics were computed for four seasons: summer (June–August), autumn (September–November), winter (December–February) and spring (March–May). These were compared to the 4 mg/L ecological threshold set by the Swiss Waters Protection Ordinance (OEaux) (Swiss Confederation, 2023).

4. Results and Interpretations

4.1 Baseline: 2024-2025 campaigns

The 2024 mooring record provides the first characterisation of Lake Taney's annual thermal cycle. From late March through April, the water column was nearly isothermal at around 4-6°C, consistent with spring mixing following winter ice cover. Pronounced summer stratification developed from late May to early June, with surface temperatures reaching a maximum of 22.5°C while the hypolimnion remained stable at approximately 6°C below 20m depth, confirming a cold, persistent bottom layer throughout the stratified season. Mixing resumed gradually from mid-September, with the water column returning to near-isothermal conditions (7-9°C) by November, consistent with a dimictic regime, two mixing events per year separated by summer stratification and winter inverse stratification under ice (Dimictic lake, 2026). A brief sensor anomaly was recorded in early June 2024, likely a logger malfunction, but does not affect the overall interpretation.

The partial 2025 record (late April to early June) captured stratification onset under warmer conditions than the equivalent 2024 period: surface temperatures rose from ~9-10°C in late April to ~16°C by early June, while temperatures below 15 m remained near 4-6°C. This earlier, stronger warming is consistent with the above-average spring temperatures recorded across Switzerland in April 2025 (6.7°C against a 1991-2020 average of 4.7°C) (Bickerstaffe et al., 2025).

June 2025 transects confirmed this stratification was spatially coherent: the shallow river-mouth area remained uniformly warm throughout the water column, while the deep basin showed a sharp thermocline with a well-developed, laterally homogenous hypolimnion (~6°C) below 15-17 m. Surface oxygen supersaturation (120%-145%) reflected photosynthesis and atmospheric exchange, declining below the thermocline;

by early June, deep-water oxygen saturation had already fallen to 30-40%, which is notably low for the start of the stratified season, suggesting spring mixing may not have fully reoxygenated the deep basin prior to stratification onset.

4.2 Results of the 2026 Campaign

This section presents the analysis of the data collected during the June 2025 - June 2026 campaign. It details the spatio-temporal variations of the measured physico-chemical parameters, highlighting the observed trends and any potential anomalies.

4.2.1 Meteorological conditions

Lake Taney sits in the Valais Rhône Valley climate zone. Sheltered from moist air masses, this inner Alpine receives 550-700 mm of precipitation annually, with temperature decreasing with elevation from the valley floor toward the surrounding pre-Alps and Alps.

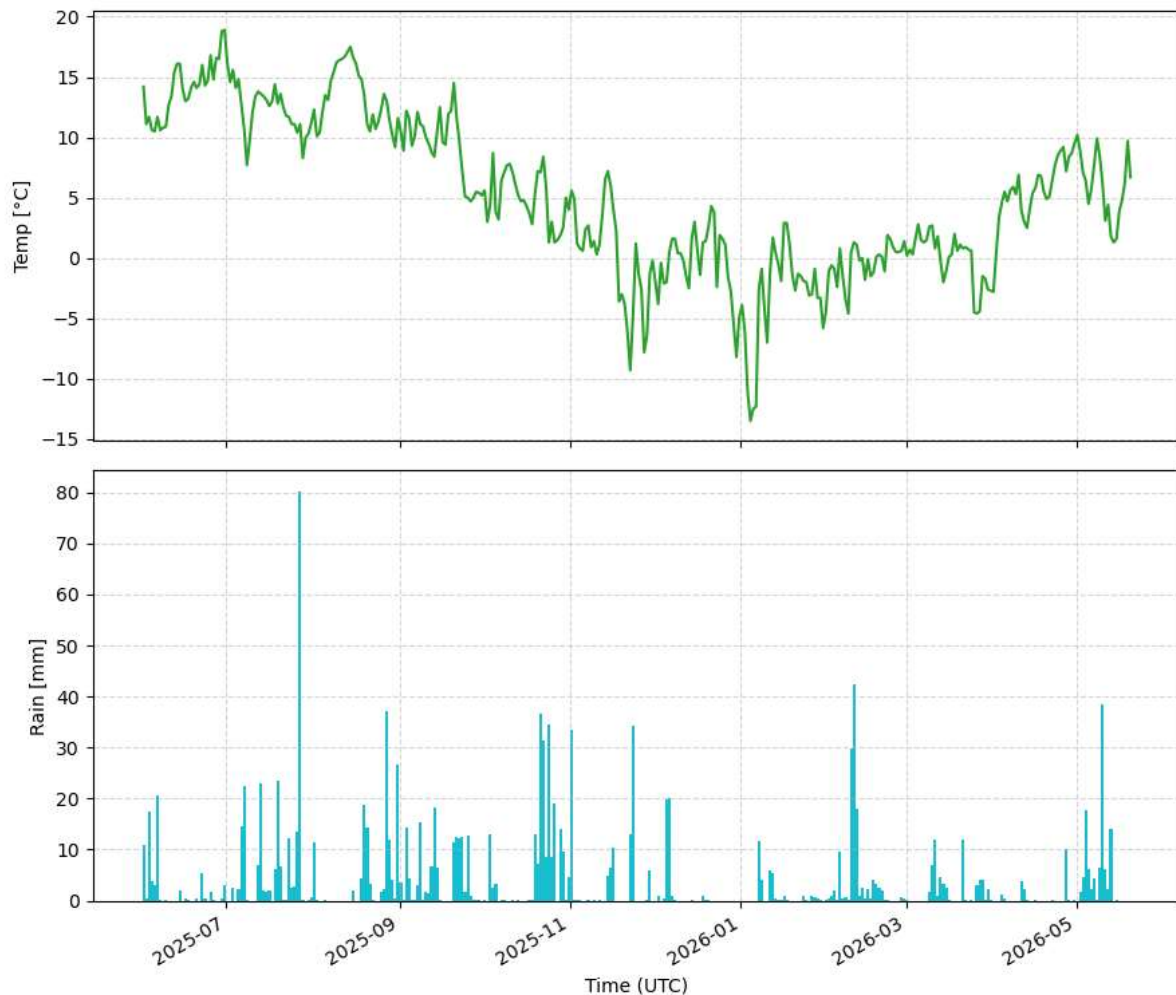


Figure 4: Air temperature (green) and precipitation (blue), Lake Taney, June 2025 - May 2026.

Air temperature followed a clear seasonal cycle, ranging from $-13.5\text{ }^{\circ}\text{C}$ to $18.9\text{ }^{\circ}\text{C}$ (mean $5.1 \pm 6.5\text{ }^{\circ}\text{C}$), with 77 daily observations below freezing in winter. Precipitation was irregular occurring in short episodic events; total recorded precipitation reached 1271.7 mm over the period (mean 3.6 mm day^{-1}), with a single major peak of 80.3 mm on 27 July 2025. Overall, these conditions describe a cold mountain climate with strong seasonal temperature contrast and episodic precipitation.

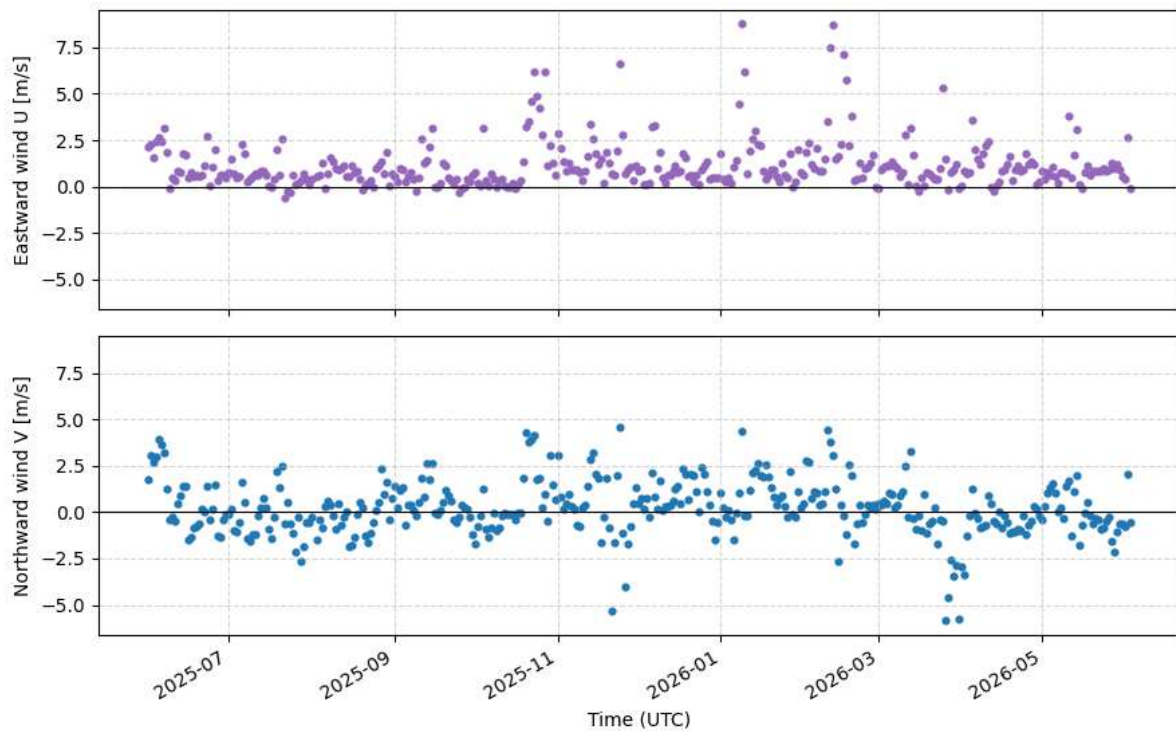


Figure 5: Eastward (U; purple) and northward (V; blue) wind components., Lake Taney, June 2025-May 2026. Positive U indicates wind directed eastward.

Given the lake and valley's east-west orientation, wind was examined for topographic channelling effects. The eastward component was predominantly positive (70% of hourly records), ranging from -6.49 to 11.72 m s^{-1} (mean $1.17 \pm 1.95\text{ m s}^{-1}$), suggesting a dominant eastward flow along the valley axis toward the Rhône Valley plain. The northward component was more balanced (56% positive, 44% negative; range -7.96 to 8.84 m s^{-1} , mean $0.23 \pm 1.88\text{ m s}^{-1}$), indicating north-south movement without a persistent direction. Overall wind speed ranged 0.02 - 13.94 m s^{-1} (mean $2.33 \pm 1.82\text{ m s}^{-1}$), a pattern consistent with topographic channelling of airflow along the valley axis (Kossmann & Sturman, 2003).

4.2.2 Hydrology

Water-level changes were derived from HOBO pressure logger measurements and used to correct sensor depths in the thermal reconstruction, linking lake-level fluctuations to the interpretation of temperature and oxygen data.

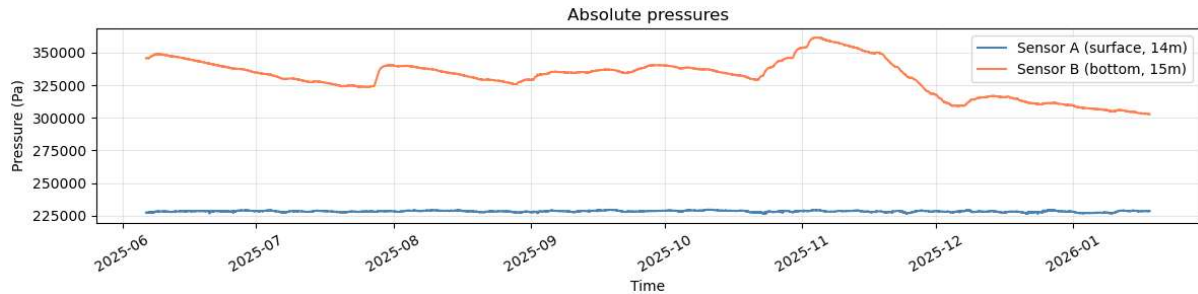


Figure 6. Absolute pressure recorded by two submerged HOBO sensors: Sensor A at 14 m depth and Sensor B positioned 5.4 m above the lake bottom, June 2025–January 2026, 15-minute resolution.

Sensor A, positioned higher on the mooring, showed a relatively stable pressure signal with low variability. Sensor B, located deeper in the water column, recorded consistently higher absolute pressure (~305,000–325,000 Pa). Pressure rose abruptly at the end of July, coinciding with the major precipitation peak (Figure 4), then increased sharply to a maximum of ~360,000 Pa in late October–early November 2025, representing the largest water-level increase in the available record. Pressure declined progressively through December 2025 and January 2026, reaching the lowest values of the deployment period.

Water-level variation was derived from the hydrostatic pressure relationship. In a water column, pressure increases with depth according to:

$$P = \rho gh$$

where P is pressure (Pa), ρ is water density (kg/m^3), g is gravitational acceleration (m/s^2), and h is water-column height (m). The relative water-level variation was calculated from the pressure difference between Sensor B and Sensor A, expressed relative to the mean pressure difference over the available record:

$$\Delta h(t) = [(P_B - P_A)(t) - \text{mean}(P_B - P_A)] / (\rho g)$$

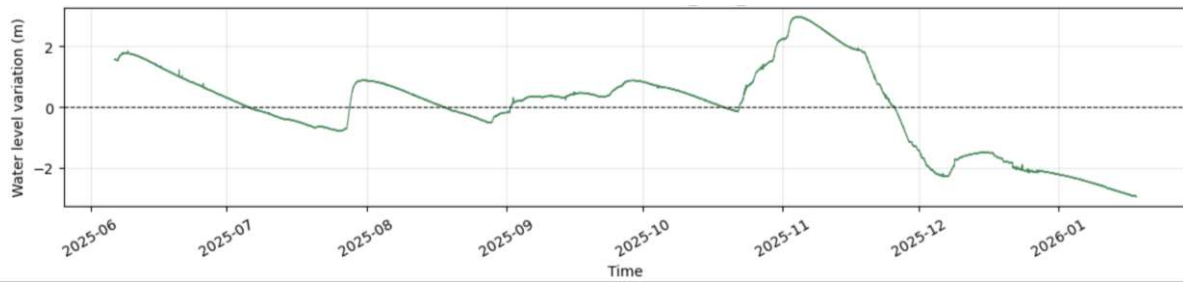


Figure 7: Water-level variation derived from the pressure difference between Sensors B and A, expressed relative to the mean pressure difference over the available record, June 2025–January 2026.

The reconstructed water-level variation ranged from -2.95 m to $+2.98$ m. Positive values indicates periods when the water level was above the time-averaged mean, while negative values indicate periods below it.

A major recharge event between October and November 2025 raised the water level to nearly $+3.0$ m above the mean, likely driven by precipitation, catchment runoff, and/or hydropower regulation. This increase coincided closely with the onset of autumn mixing and the associated dissolved-oxygen peak described in a later section. From late November onward, water level decreased strongly and continuously through January 2026, reaching the lowest values of the available pressure record.

4.2.3 CTD profiles

The June 4th, 2026, CTD profiles results are presented as temperature, measured conductivity, conductivity corrected at 20°C , salinity, dissolved oxygen concentration, and oxygen saturation as a function of depth.

The point 9 from transect 2 (12:12 in lime green) showed anomalously high conductivity and salinity throughout the entire profile. This outlier is likely due to a displacement of the CTD probe during measurement and was excluded from all subsequent analysis. (Figure 8).

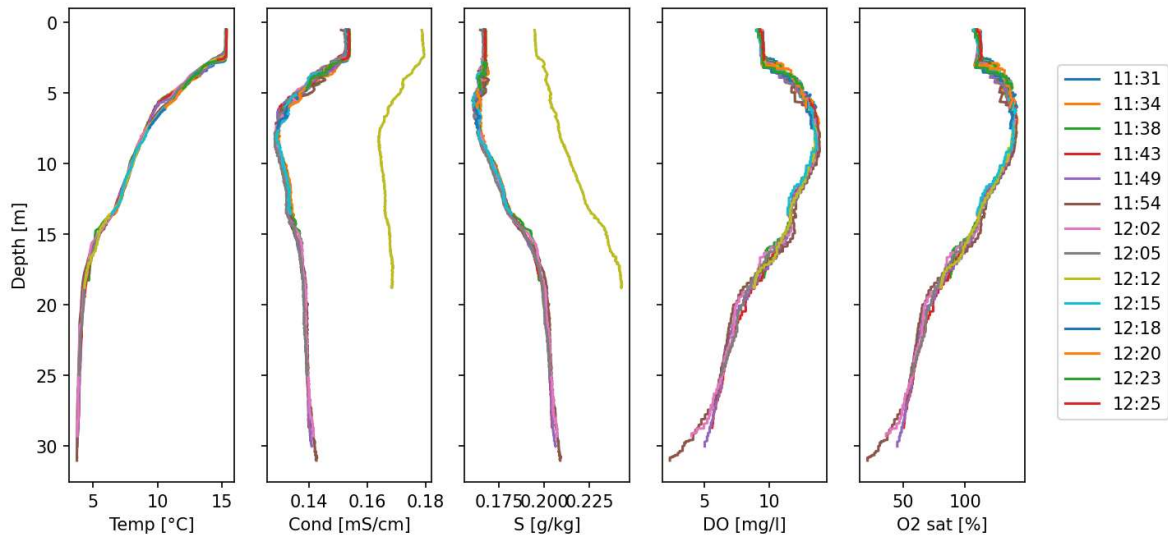


Figure 8: Vertical profiles of temperature (Temp, °C), conductivity (Cond, mS/cm), salinity (S, g/kg), dissolved oxygen (DO, mg/L) and oxygen saturation (O₂ sat, %) for Transect 2, 4 June 2026

Temperature

Overall, the profiles are nearly identical across all stations of both transects, indicating the lake is horizontally well-mixed for temperature (Figure 9).

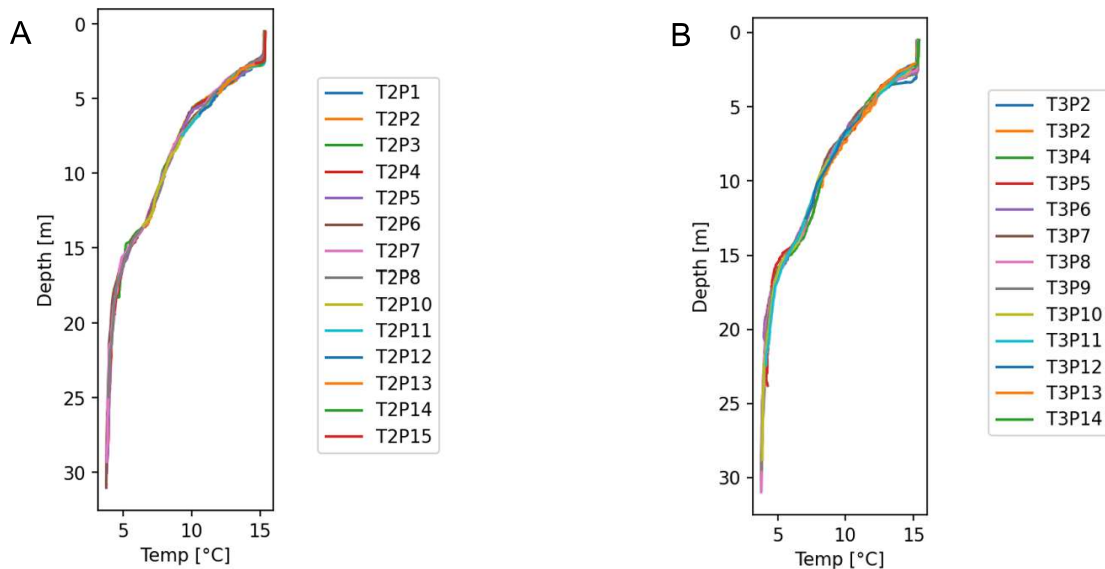


Figure 9: Vertical temperature profiles (Temp, °C) for (A) Transect T2 and (B) Transect T3, 4 June 2026.

Both transects show thermal stratification at the time of sampling: surface waters near 15 °C drops sharply across the metalimnion (3–14 m) to ~7 °C, stabilising at 3.75–4 °C toward the bottom. Cross-sections confirm consistent layering along both transects (Figure 10).

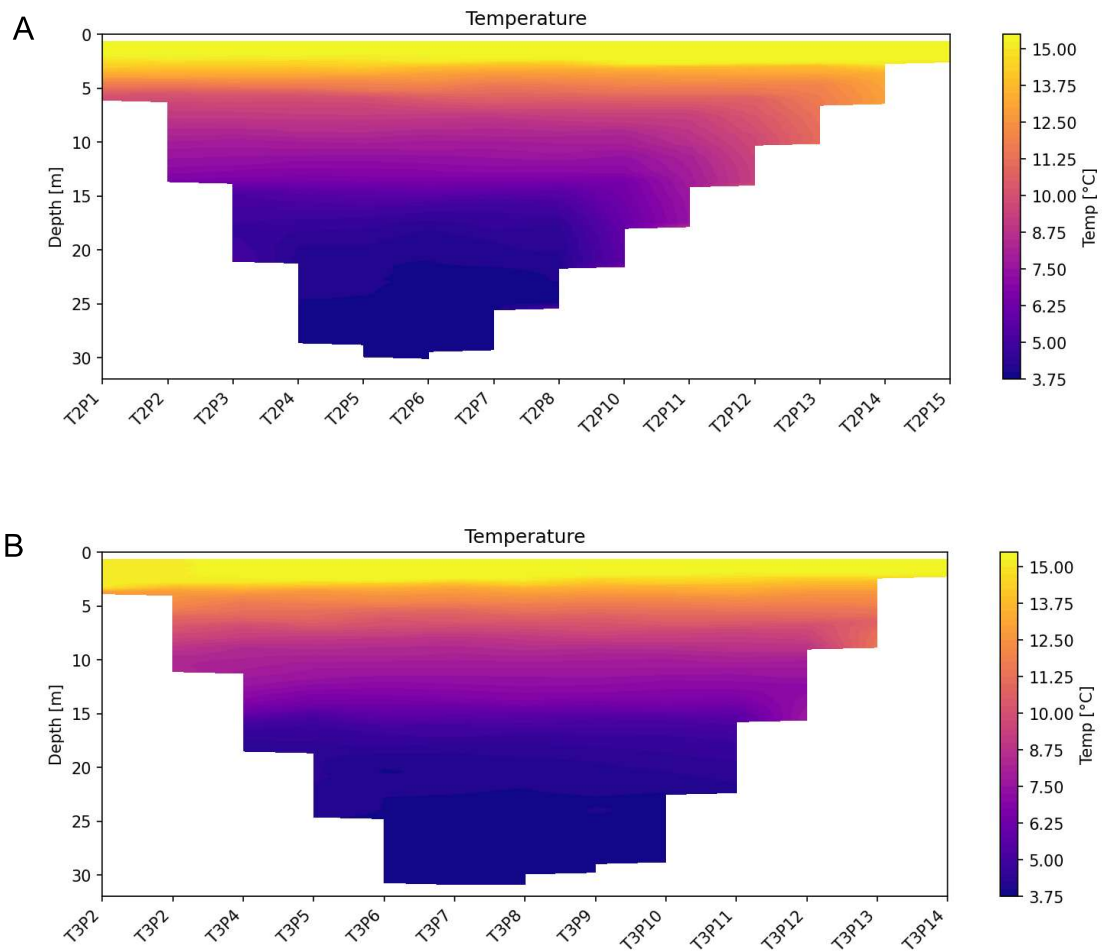


Figure 10: Time–depth cross-section of temperature, (A) Transect T2, (B) Transect 3.

This structure is characteristic of a dimictic lake under summer stratification, with the metalimnion separating the warm epilimnion (surface to ~3 m) from the cold hypolimnion (~14 m to the bottom at ~31 m) (Wells and Troy, 2022).

Conductivity

Electrical reflects dissolved ion concentration and increases with temperature; measured values are standardised to a 20 °C reference to isolate the ion-content signal (Hayashi, 2004).

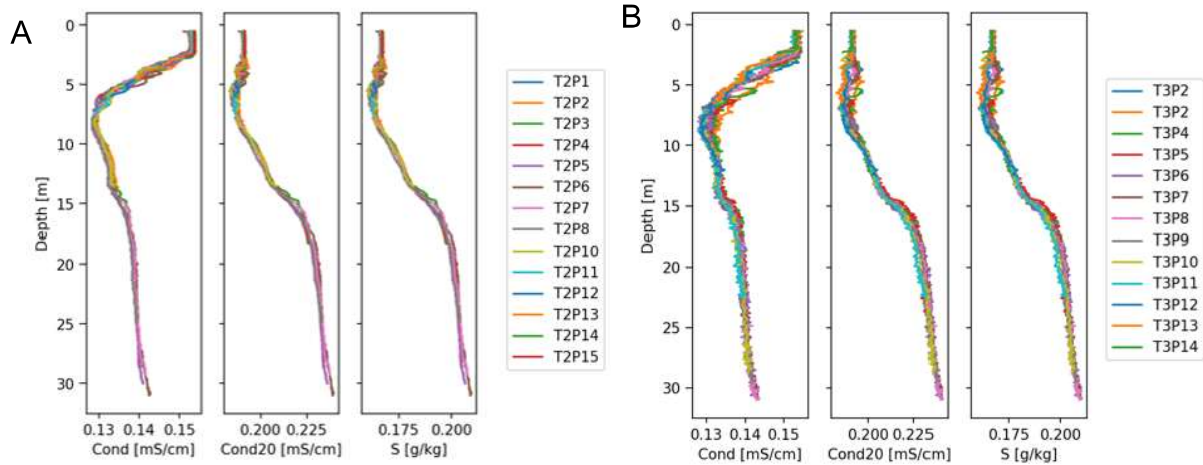


Figure 11: Vertical profiles of conductivity (Cond, mS/cm) , conductivity corrected at 20°C (Cond20, mS/cm), salinity (S, g/kg) for **(A)**Transect T2, **(B)** Transect T3, 4 June 2026.

Profiles are similar across all stations within both transects, further supporting horizontal homogeneity on the sampling date. Measured conductivity ranged 0.13-0.16 mS/cm, highest values in the epilimnion, decreasing rapidly between 3 to 7 m to a minimum, then stabilising near 0.14 mS/cm in the hypolimnion below 14 m, mirroring the temperature-defined layering (Figure 9) and indicating a strong correlation between the two variables.

In contrast, Cond20 increased steadily with depth (from ~0.19 to 0.24 mS/cm), without the distinct layering seen in raw conductivity, and salinity followed the same trend (0.16 to 0.21 g/kg), as it is computed from Cond20. This depth-increasing pattern of Cond20 and salinity indicates an enrichment of dissolved ions in the hypolimnion, characteristic of stratified lakes in summer, where limited vertical mixing allows ions released by decomposing organic matter and sediment-water exchange to accumulate at depth (Rimmer et al., 2005).

Oxygen

The vertical profiles of dissolved oxygen (DO) concentration and oxygen saturation show a very similar structure in both transects (Figure 12), again confirming horizontal homogeneity.

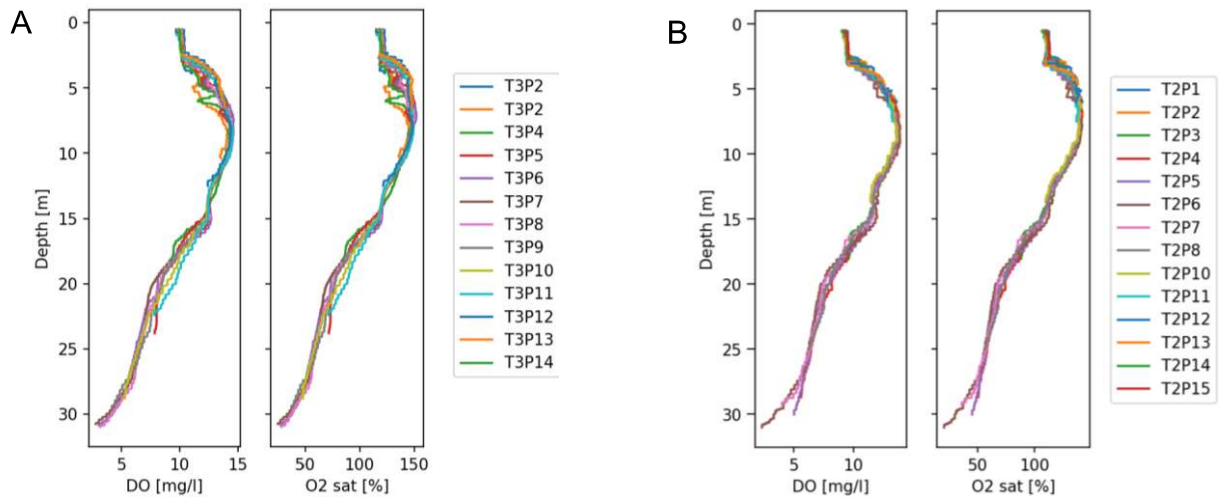


Figure 12: Vertical profiles of dissolved oxygen (DO, mg/L) and oxygen saturation (O_2 sat, %) for **(A)** Transect T2, **(B)** Transect T3, 4 June 2026.

At the surface, O_2 saturation approaches atmospheric equilibrium (~100%) with concentrations near 10 mg/L, consistent with well-mixed surface waters continuously exchanging oxygen with the atmosphere (Staehr et al., 2012). The highest concentrations are observed in the upper metalimnion (maximum of 13.5 mg/L for T2 and 14 mg/L for T3; saturation 130–145%), most likely reflecting intense photosynthetic activity in the photic zone (Figures 12 and 13)(Staehr et al., 2010), with thermal stratification limiting vertical mixing and allowing oxygen produced there to accumulate. The June sampling coincides with the spring phytoplankton bloom, and abundant aquatic vegetation observed in the field supports high primary productivity during this period.

In the hypolimnion, insufficient light limits photosynthesis while microbial respiration and decomposition continue to consume oxygen, and stratification restricts the transport of oxygen-rich water from the surface to the bottom of the lake, resulting in a progressive decrease in dissolved oxygen with depth. (Staehr et al., 2010). Minimum concentrations near the bottom reached 2.25 mg L^{-1} (T2) and 2.75 mg/L (T3) (Figure 13). Dissolved oxygen fell below the 4 mg/L hypoxia limit at stations T2P6, T3P6, T3P7 and T3P8, where only stress-tolerant species could persist; concentrations approached but did not reach the 2 mg L^{-1} anoxic threshold. As expected, (Staehr et al., 2012), the hypolimnion also showed the lowest oxygen saturation, reflecting negative net ecosystem production, where respiration exceeds primary production.

These June 2026 values likely represent only the early stages of seasonal depletion, since sampling occurred at the start of the stratified period; hypolimnetic oxygen is expected to continue declining through summer as respiration persists under limited vertical exchange, a trajectory confirmed by the continuous mooring record.

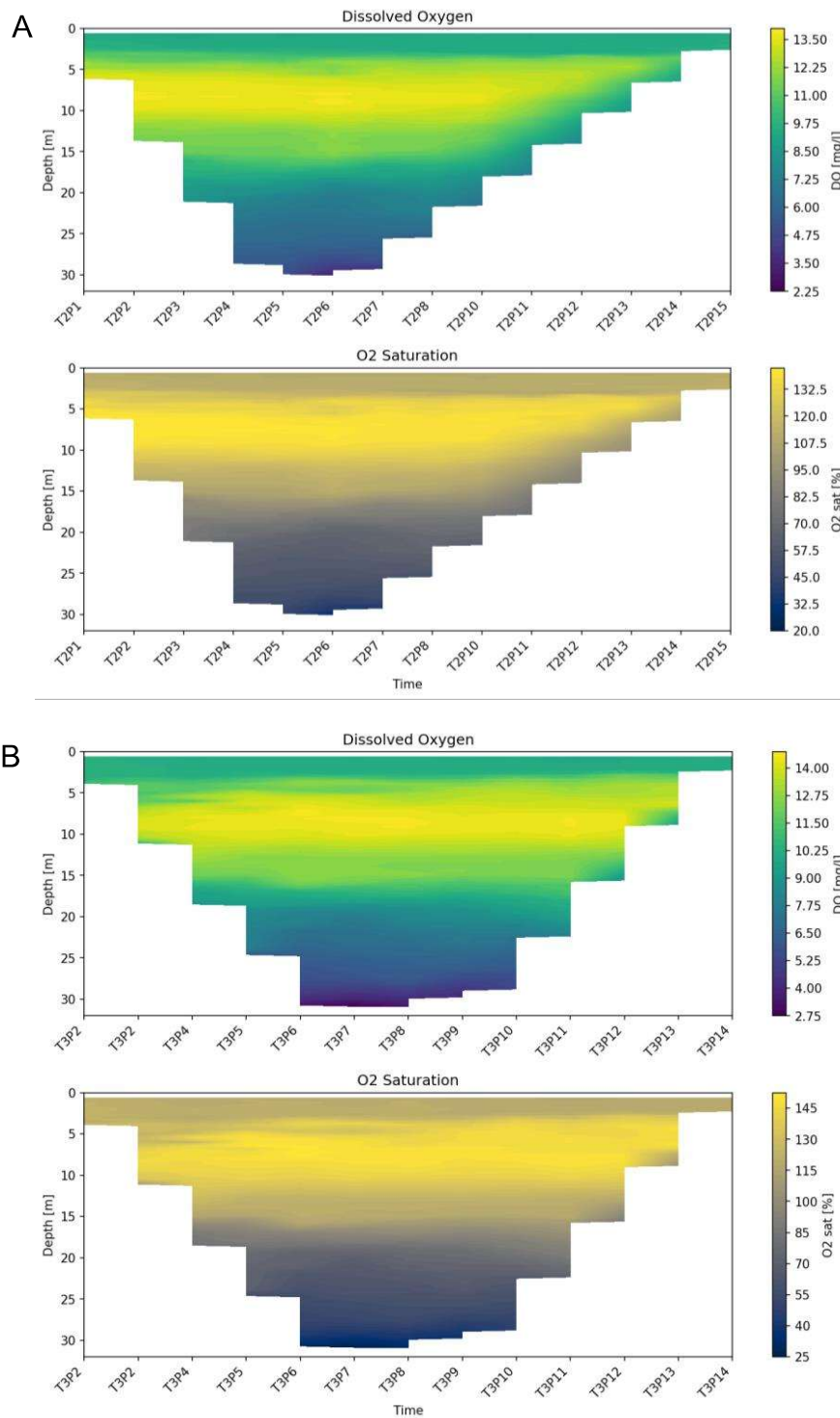


Figure 13: Time-depth cross-section of dissolved oxygen and oxygen saturation along
A) Transect 2 B) Transect 3.

4.2.4 Bathymetry

The large standard deviation reflects pronounced bathymetric variability particularly the steep littoral slopes. The highest observation frequency occurred between 30 and 35 m depth, indicating that a large proportion of the lake bottom consists of a relatively flat deep basin at ~30 m, bordered by steep shoreline slopes (Figure 14).

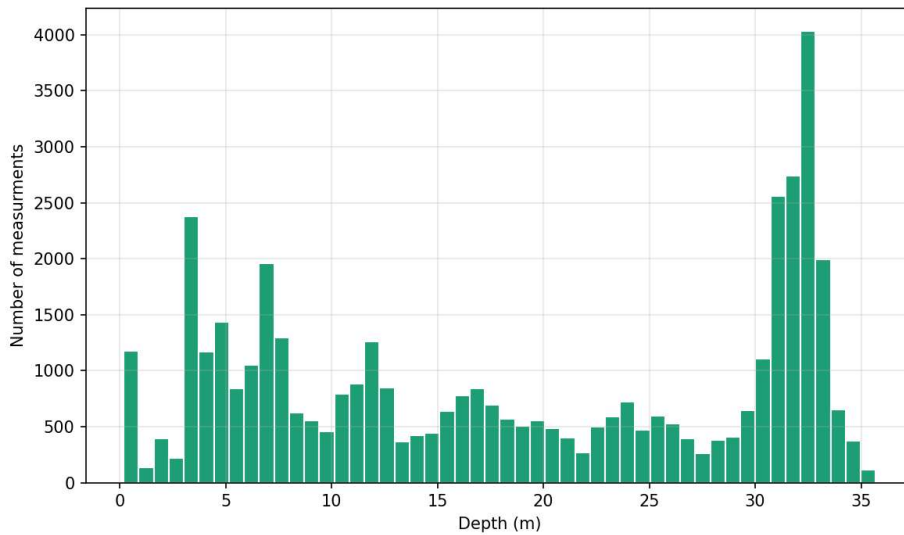


Figure 14: Frequency distribution of water depths along Transect T2, 4 June 2026 (n = 43,568). Depths ranged 0.19-35.70 m (mean = 18.74 ± 11.34 m).

Despite lateral vessel drift, the raw profile clearly identifies the deepest section (max 35.70 m), closely matching the reported 35 m maximum depth (Figure 15).

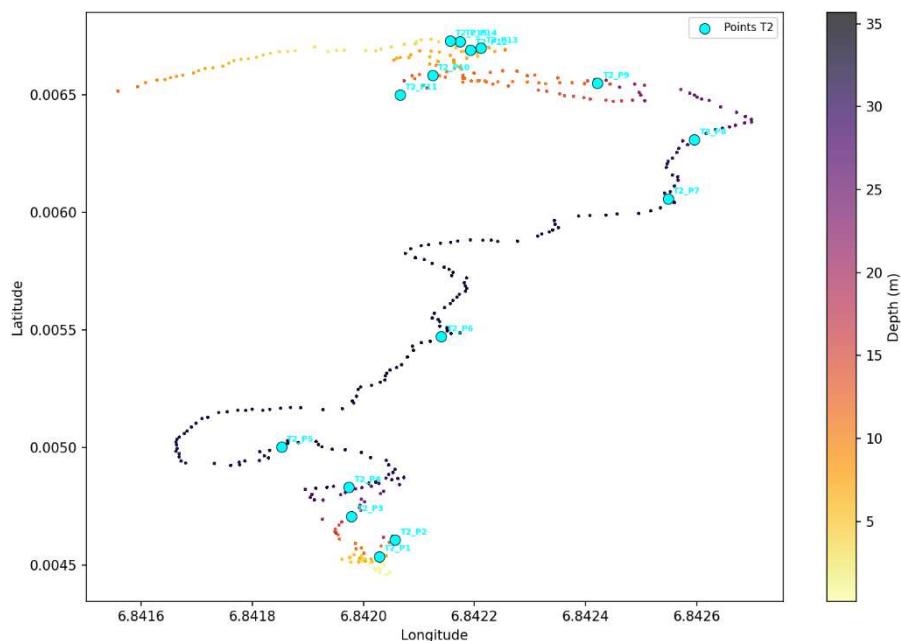


Figure 15: Raw georeferenced depth measurements along Transect T2 (WGS84), coloured by depth (m); cyan markers show the 15 CTD stations. Collected 11:27-12:30 (UTC+2), 4 June 2026 (n = 43,568).

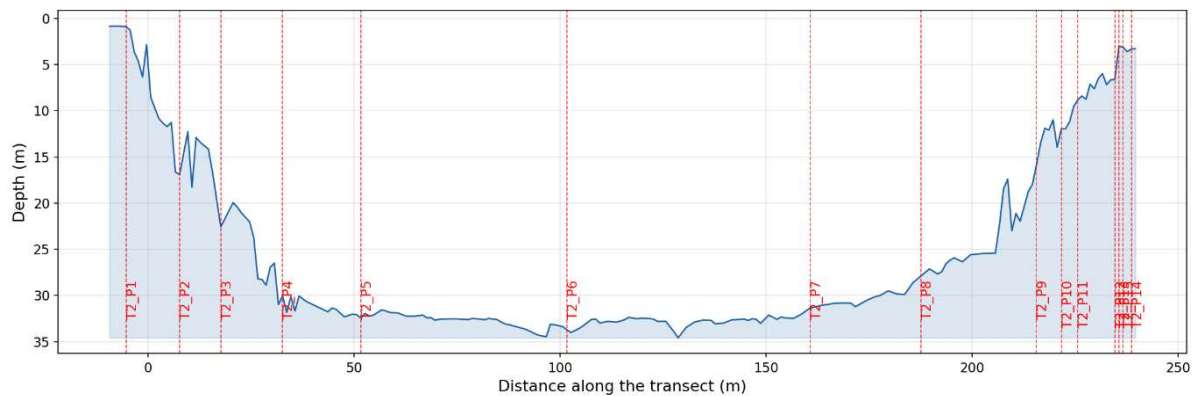


Figure 16: Spatially smoothed bathymetric profile along Transect T2, derived from orthogonal projection onto a straight reference axis with 1 m bin averaging ($n = 178$ points). Vertical red dashed lines indicate the 15 CTD stations, 4 June, 2026.

The corrected transect length (222 m) is consistent with the reported lake width (270 m). The profile in Figure 16, shows progressive deepening from the southern end (near the hydropower outflow), reaching maximum depth ~35 m from shore in the central lake, the steepening again toward the northern cliff face ~30 m from the northern shore, an overall relatively symmetrical cross-section.

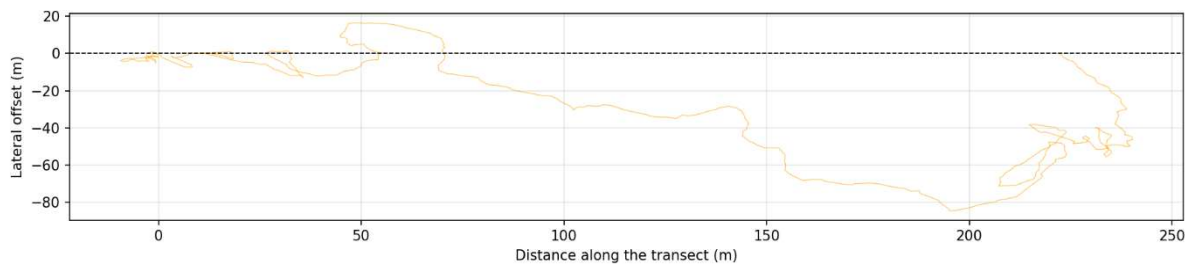


Figure 17. Lateral drift of the survey vessel relative to the straight reference axis Transect T2, 4 June, 2026. Positive (west) and negative (east) values indicate deviations to either side of the reference axis. The dashed horizontal line represents the ideal straight-line trajectory.

Maximum lateral deviation reached 84.7 m, reflecting wind- and current-driven forces on the unanchored vessel, and justifying the orthogonal-projection correction applied in previous sections, without which the raw trajectory would introduce substantial spatial redundancy and distort the bathymetric profile (Figure 17).

4.2.5 Thermal structure and meteorological forcing

Sensors range from yellow (near-surface) to blue-green (depth) (Figure 18). Stratification periods are clearly visible between July and October 2025, and again from April to May 2026, with surface temperatures much warmer than at depth. The shallowest sensor (S10) shows the strongest variability, while the deepest sensors (S2 and the MiniDOT) remain stable and largely unaffected by stratification. A pronounced disruption on event 27 July corresponds to a major storm event, examined in detail in later section.

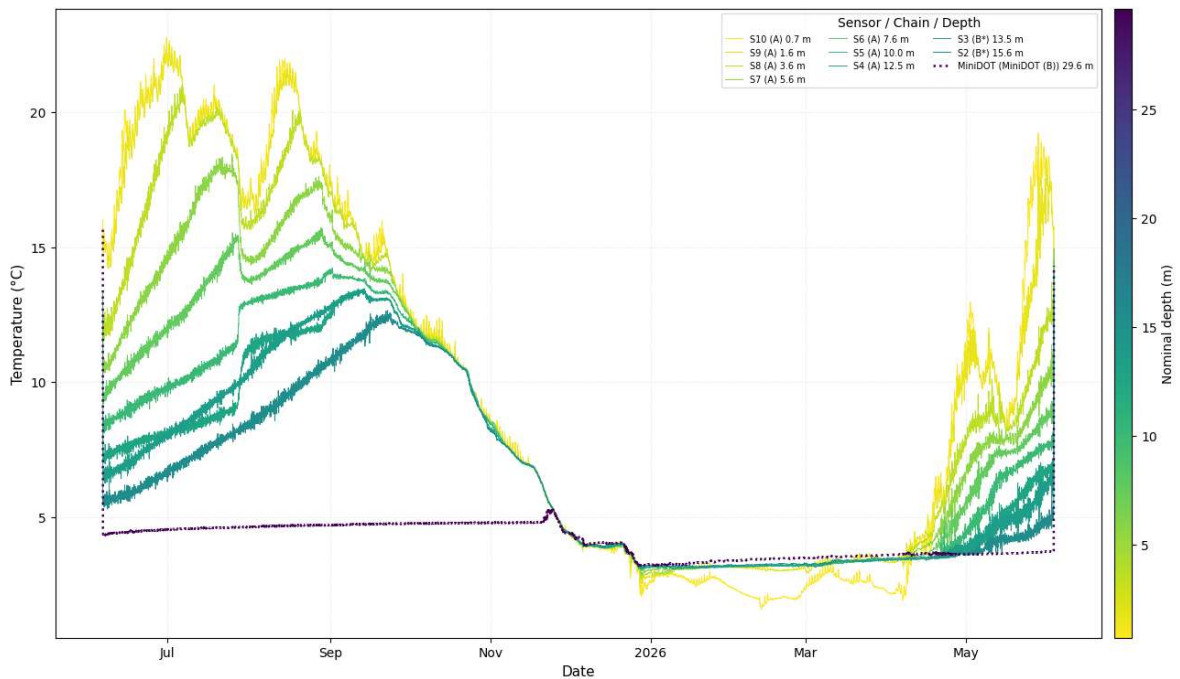


Figure 18 : Temperature time series by sensor depth, July 2025 -May 2026 (Chain A fixed; Chain B moving with depth; bottom temperature from the MiniDOT, 1.4 m above the lake bed).

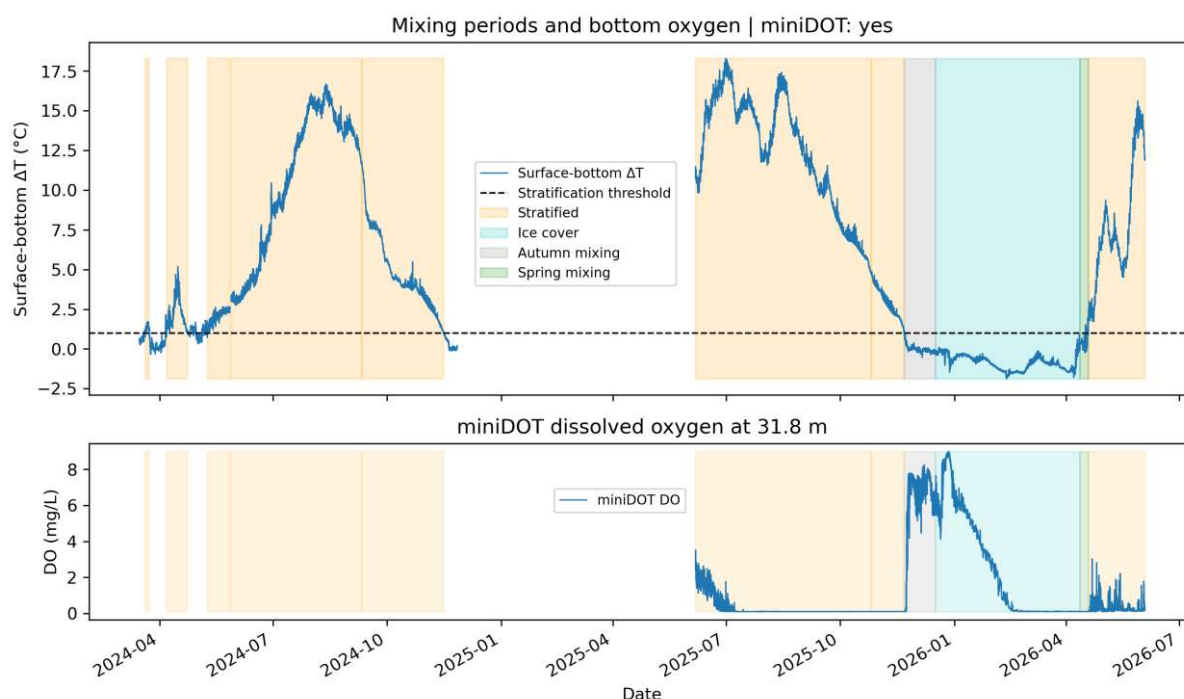


Figure 19 : Mixing periods, bottom oxygen and thermal regime, April 2024-July 2026.

The lake was strongly stratified in summer 2024 and 2025. The longest stratified period ran from June to November 2025 (~168 days), close to the ~171-day stratified period in 2024 (Table 1). Ice cover was inferred from mid-December 2025 to mid-April 2026 (~116 days), during which the temperature structure inverted (bottom slightly warmer than surface, Figure 19). Dissolved oxygen declined progressively under ice, indicating winter anoxia development, then increased after ice-off, consistent with spring mixing and reoxygenation.

Autumn mixing (~25 days) was substantially longer than spring mixing (~7 days), a pattern consistent with a dimictic regime but suggesting that late ice-off is rapidly followed by renewed surface warming and stratification before the water column can fully homogenise. With only one complete ice-covered winter in the current record, it is not yet possible to determine whether this asymmetry is typical of Lake Taney or specific to 2025–2026.

Table 1: Summary of the key stratification and mixing, Lake Taney, 2024-2026.

Period type	Start date	End date	Duration days
Stratified	20.03.2024	23. 03.2024	2.8
Stratified	06. 04.2024	23. 04.2024	16.7
Stratified	09. 05.2024	15. 11.2024	171.2
Stratified	06.06.2025	21.11.2025	168.1
Stratified	18.04.2026	03.06.2026	45.7
Ice cover	16.12.2025	11.04.2026	116.3
Autumn mixing	21.11.2025	16.12.2025	25.2
Spring mixing	11.04.2026	18.04.2026	6.9

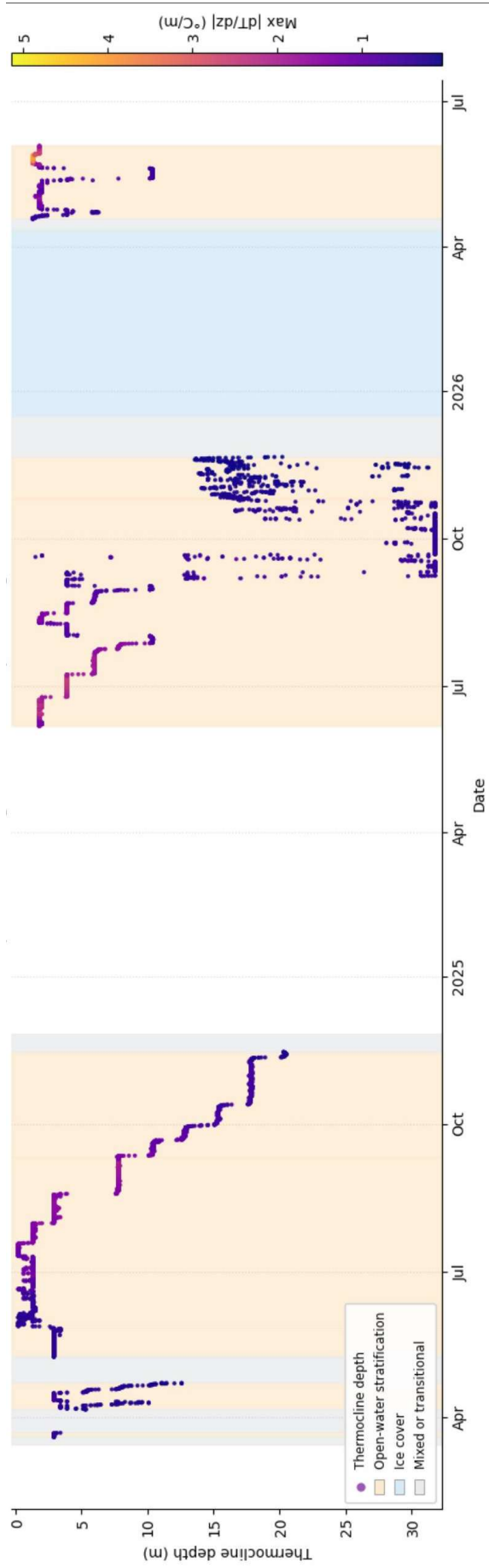


Figure 20: Thermocline depth and thermal structure of Lake Taney, June 2024-June 2026 (2024-2025 campaign, left; 2025-2026 campaign, right). Background colour indicates active regime: orange (stratification), blue (ice cover), grey (transition).

In 2024-2025, the thermocline remained at 2-3 m through spring and summer, deepening gradually to ~20 m by November, as the water column cooled and began mixing ; the gradient stayed weak throughout (dark purple, at most around 1.5 °C/m).

In 2025-2026, the thermocline started shallow (~2 m) in June, deepened through July-August, to 10-12 m by early autumn, with a notably stronger gradient (pink-orange) than previous year. Between September and October 2025, thermocline-depth estimates scatter widely (15–32.8 m), reflecting a sensor coverage gap rather than a physical signal: the widening spacing of sensors toward depth, combined with a 14 m gap between S2 and the MiniDOT with no intermediate measurement, forces the interpolation algorithm to locate the thermocline across a 14 m blind zone using only two anchor points. The underlying deepening trend is real, but the exact depth cannot be resolved precisely during this window. Ice cover (~December 2025–April 2026) precludes thermocline computation, as expected under inverse stratification with no warm surface layer (Wetzel, 2001). Stratification rebuilt in spring 2026, returning the thermocline near the surface by May-June.

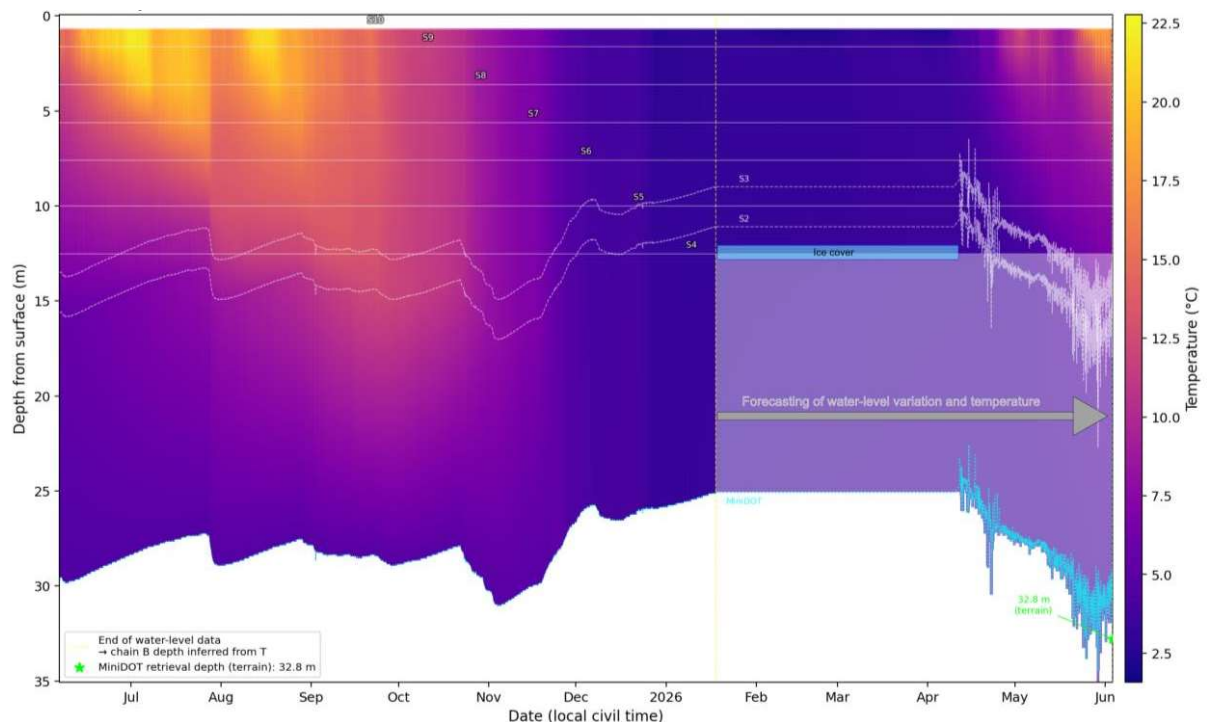


Figure 21: Variation of the depth with the thermal Structure 2025-2026, with assumptions for the missing pressure data.

This figure represents the lake depth referenced from the surface. Sensors S10 to S4 are mounted on chain A and therefore follow variations in the lake surface level. In contrast, sensors S2 and S3 are attached to chain B and allow the reconstruction of water level changes over time. The MiniDOT sensor, located 14 m below S2 on chain B, provides near-bottom temperature measurements.

An increase in water level is observed in August and, more markedly, in November, likely due to precipitation events. In December and January, precipitation is lower, and

we hypothesize that hydropower extraction is stronger during this period, resulting in a decrease in water level. From 18 January 2026, onward, pressure data are no longer available, preventing direct estimation of lake depth. We therefore extrapolated depths assuming constant water level during the ice-cover period (December 16th, 2025, to April 10th, 2026).

The temperature field was reconstructed on a 0.1 m depth grid at each hourly timestep, following the Chain B correction to match the measured maximum depth of 32.8 m recorded during mooring retrieval on June 3, 2026. Future observations (e.g., in 2027) will be needed to determine whether these patterns are consistent between years or specific to this dataset.

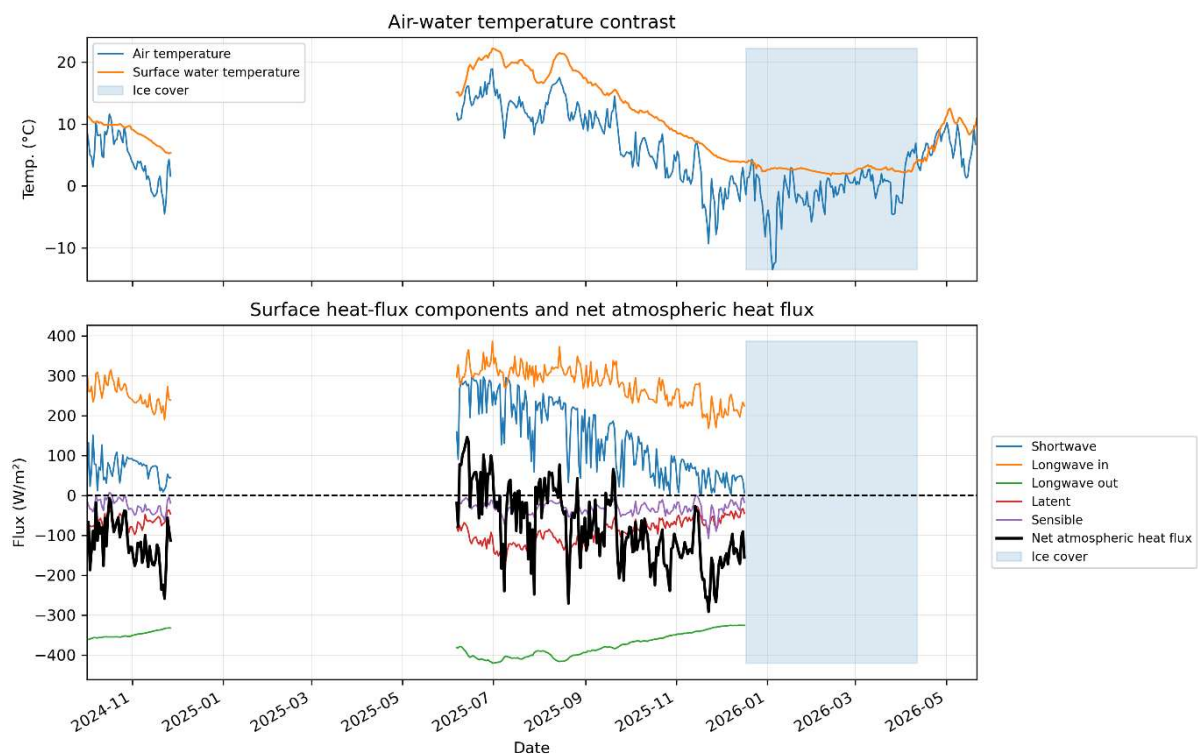


Figure 22: Air-water temperature contrast and surface heat-flux components, November 2024-June 2026. Upper panel: air and surface-water temperature. Lower panel: shortwave, longwave-in, longwave-out, latent heat, sensible heat, and net atmospheric heat flux. Blue shading: ice-cover period.

Atmospheric heat-flux components were computed to describe the exchange of energy between the lake surface and the atmosphere (Figure 22). The upper panel shows the contrast between air temperature and surface-water temperature, while the lower panel shows the different heat-flux components and the resulting net atmospheric heat flux. After summer, the decrease in air temperature is associated with a progressive decline in net heat flux, indicating a gradual loss of heat from the lake surface. During the ice-covered period, direct atmospheric heat exchange is strongly reduced, as the ice cover limits the interaction between the water surface and the atmosphere.

4.2.6 Dissolved oxygen

Bottom dissolved oxygen (31.8 m) was recorded continuously over 362 days (6 June 2025 - 3 June 2026). Under the Swiss Waters Protection Ordinance (OEaux), a minimum concentration of 4 mg/L is required to maintain acceptable ecological conditions for aquatic organisms; this threshold is the primary reference throughout this section. The sensor recorded a minimum of 0.09 mg/L, a maximum of 9.02 mg/L, and a global mean of 1.40 mg/L, bottom waters remained far below the legal standard for most of the year. Bottom temperature was remarkably stable (3.15–5.38°C, mean 4.12°C), confirming a cold, isolated hypolimnion decoupled from seasonal surface warming.

Seasonal dynamics

The time series reveals a clear seasonal cycle structured around two mixing events and two periods of prolonged anoxia (Figure 23).

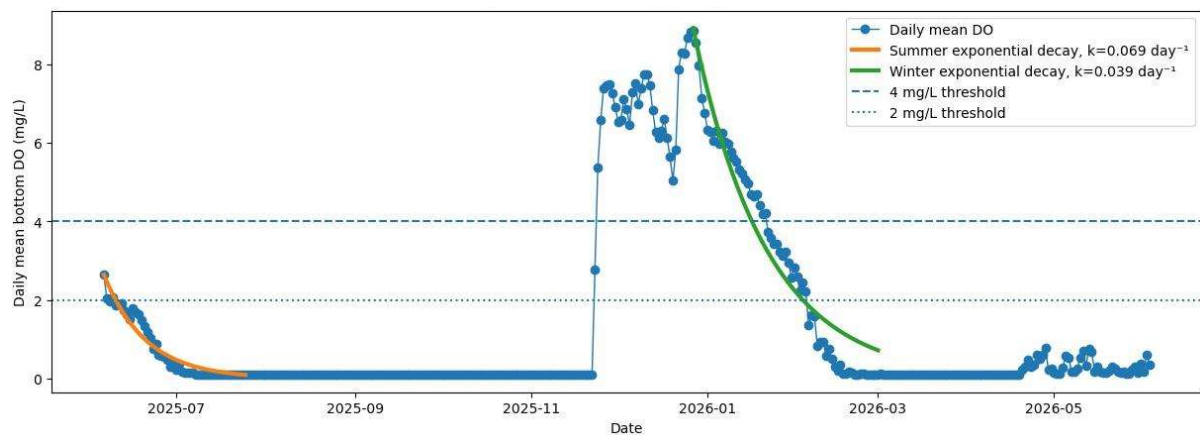


Figure 23. Daily mean dissolved oxygen concentration at 31.8 m depth, June 2025–June 2026. Dashed blue line: 4 mg/L threshold (OEaux); dotted blue line : 2 mg/L threshold. Orange and green curves : exponential decay fits for summer ($k = 0.069 \text{ day}^{-1}$) and winter under-ice periods ($k = 0.039 \text{ day}^{-1}$), following Perga et al. (2023). Gaps correspond to periods without data.

From deployment (early June 2025) through late November, bottom DO declined rapidly from ~2.5 mg/L toward near-zero, remaining there for roughly five months. DO was already below 4 mg/L at deployment (~2.5 mg/L), indicating summer deoxygenation had begun before monitoring started, consistent with the onset of summer stratification, which isolates the hypolimnion from surface exchange and allows progressive oxygen consumption by sediment respiration and organic matter decomposition (Jane et al., 2023).

In December 2025, DO rose sharply to 9.02 mg/L, marking winter mixing onset: as surface water cooled and density stratification broke down, oxygenated surface water reached the deep layers, the only full bottom reoxygenation recorded during the monitoring period, and corresponding to the autumn mixing event identified (Table 1).

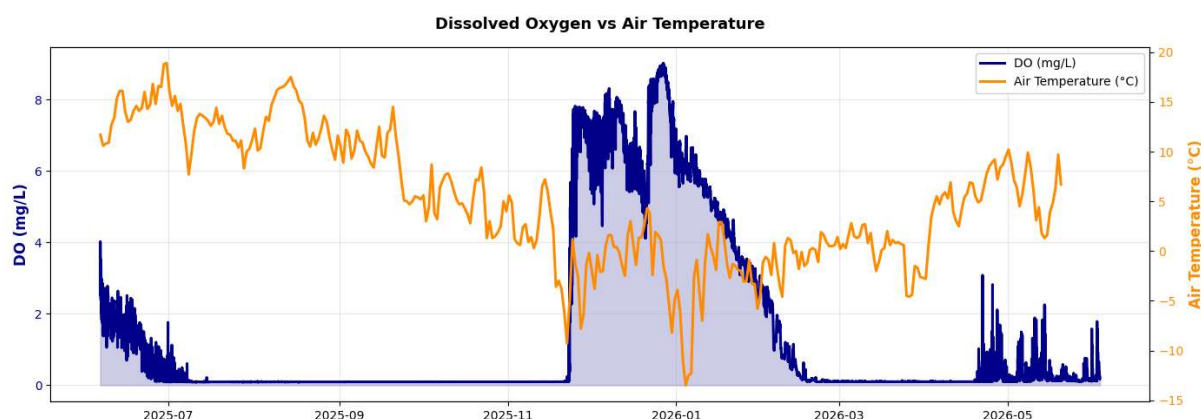


Figure 24: Daily mean dissolved oxygen concentration at 31.8 m depth (left axis, blue) and air temperature recorded at the local meteorological station (right axis, orange), Lake Taney, June 2025 - June 2026. The December 2025 DO peak coincides with a sustained drop in air temperature below zero, consistent with surface cooling driving convective mixing prior to ice formation

Following this peak, DO declined steadily through the ice-covered winter period (late December 2025 to mid-April 2026; Table 1). Underice, the lake surface is sealed from atmospheric exchange, and oxygen is consumed by biological and sediment processes without replenishment.

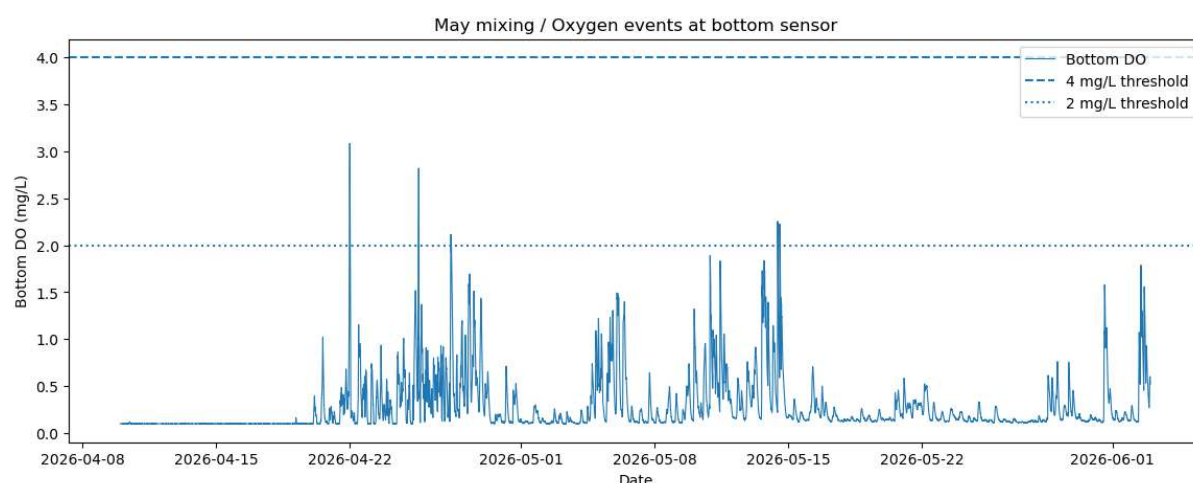


Figure 25: Bottom dissolved oxygen concentration at 31.8 m depth during the spring transition period, Lake Taney, April 8 – June 3, 2026. Short-lived spikes remain consistently below the 4 mg/L threshold (dashed blue line) and the 2 mg/L threshold (dotted blue line), indicating incomplete spring mixing.

Around late April and May 2026, intermittent increases in bottom DO are visible in the time series, coinciding with the spring mixing period identified from the thermal record (Figure 19, Section 2.5). A zoomed view of this period (Figure 25) shows that these fluctuations consist of sharp, short-lived spikes reaching up to approximately 3 mg/L, remaining consistently below the 4 mg/L threshold and returning immediately to near-zero values. Spring mixing at Lake Taney is therefore incomplete: the water column

partially homogenizes following ice-off but circulation is insufficient to deliver and sustain enough oxygen at 31.8 m depth to meet ecological requirements.

Two processes may explain the structure of these spikes. The snowmelt and ice-off period may generate episodic cold, dense surface inflows that can drive brief convective pulses to depth, consistent with the timing of the signals observed in mid-to-late April. However, given the magnitude and abruptness of these fluctuations, a second hypothesis appears more plausible: the sensor may have been partially buried by sediment resuspension during the spring transition, with episodic disturbances temporarily exposing the sensor to slightly less depleted interstitial or near-bottom water. This interpretation is supported by the irregular, spike-like character of the signal, which is inconsistent with the smoother oxygen increase that would be expected from gradual mixing. The ambiguity between these two interpretations cannot be fully resolved with the current single-sensor configuration at the bottom. This limitation directly motivates the placement of a second dissolved oxygen sensor at an intermediate depth during the next deployment, which would allow the vertical extent of any mixing signal to be tracked and distinguish a genuine water column response from a near-sediment artefact.

Legal thresholds and ecological status

At 31.8 m depth, the 4 mg/L threshold was not met for 83.7% of the monitoring period (303 of 362 days), and the 2 mg/L threshold for 78.6% (285 days). The longest continuous near-anoxic episode lasted 147 days, from June 29th to November 22nd 2025. Table 2 details exceedance by season.

Table 2. Days below the 4 mg/L dissolved oxygen threshold (OEaux) at 31.8 m depth, by season, Lake Taney.

Season	Total days	Days below 4 mg/L	% of season
Summer 2025 (Jun–Aug)	87*	87	100 %
Fall 2025 (Sep–Nov)	91	84	92.3 %
Winter 2025/26 (Dec–Feb)	90	38	42.2 %
Spring 2026 (Mar–May)	92	92	100 %

*Sensor deployed June 6, 2025; summer 2025 record covers 87 days only.

These results indicate that bottom waters at 31.8 m fail to meet the legal oxygen threshold for most of the year. The lake is effectively anoxic at depth throughout summer and spring, with only a partial respite during winter in the weeks following the December mixing event. Whether this pattern constitutes a formal non-compliance under OEaux depends on the applicable water body classification and whether the standard applies to the full water column or only to defined reference depths, a point that warrants clarification with cantonal authorities.

Oxygen decay rates

To characterize the rate of oxygen depletion during stratified and ice-covered periods, an exponential decay model was fitted following Perga et al. (2023):

$$DO(t) = DO_{init} \times \exp(-k \times t)$$

Two decay rates were estimated (Figure 23). For the summer period, $k = 0.069 \text{ day}^{-1}$, starting from sensor deployment in June 2025 when DO at the bottom was already approximately 2.5 mg/L. This estimate should be interpreted with caution: because the sensor was deployed after stratification had already begun, the true onset of summer deoxygenation was not captured, and this value likely reflects only the final stage of an already advanced depletion process. For the winter under-ice period, $k = 0.039 \text{ day}^{-1}$, starting from the post-mixing peak of 9.02 mg/L recorded on December 26, 2025. This estimate is more robust, as both the initial concentration and the start of the depletion period are well constrained.

Applied to the winter initial value, the model predicts that bottom DO would fall below the 4 mg/L threshold after approximately 20.4 days of ice cover. The summer decay rate is nearly twice the winter rate, consistent with higher sediment oxygen demand during the stratified season. Perga et al. (2023) demonstrate that across 14 alpine lakes, the decay rate under ice is the primary predictor of hypoxia duration, more so than either the initial DO concentration or the length of the ice-cover period, and that this rate is primarily controlled by physical processes, specifically near-bed warming driven by sediment heat release, rather than by biogeochemical proxies for productivity. The role of organic matter decomposition in driving the summer rate at Lake Taney therefore remains to be confirmed, and the elevated organic load hypothesized from the lake's restricted flushing and productive littoral zone cannot be confirmed without sediment or productivity data.

The winter decay rate estimated for Lake Taney ($k = 0.039 \text{ day}^{-1}$) falls within the upper range of values reported by Perga et al. (2023) across 14 alpine lakes ($k_{exp} = 0.0016$ to 0.0626 day^{-1}), suggesting a relatively rapid under-ice oxygen depletion compared to similar systems. This is consistent with a lake where near-bed thermal conditions and sediment heat release sustain an elevated physical oxygen demand throughout the ice-covered period.

Tracking the evolution of k from year to year will therefore be a key monitoring metric for Lake Taney, independent of interannual variability in ice phenology or mixing intensity and will constitute a valuable baseline for future campaigns.

4.2.7 Focus on big storm weather event

Figure 26 shows a zoom of the temperature time series during the storm event of 27 July. The storm produced a shallow mixing event, visible through the cooling of the upper sensors and the simultaneous warming of S5. Since this warming is not clearly observed at Section 4, the mixing depth can be estimated between 10 and 12.5 m.

The water-level change described in Figure 7 can also be inferred from the progressive shift of the deepest sensor of chain A relative to the upper sensor of chain B, suggesting a change in their relative depth position during the event.

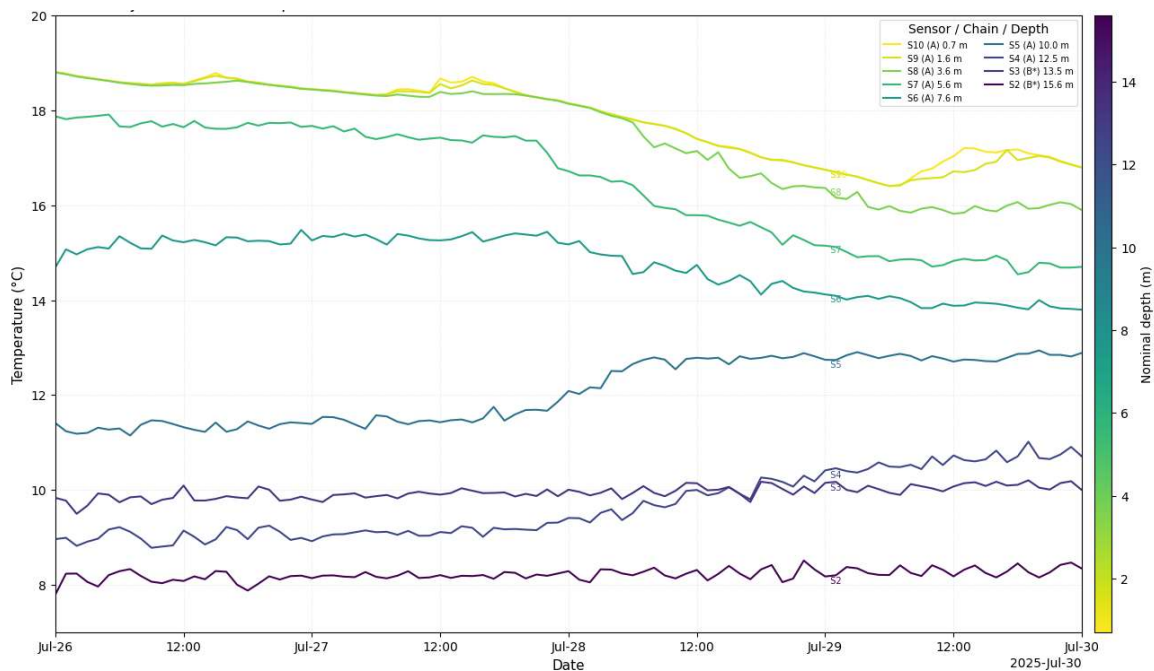


Figure 26: Temperature time series by depth from July 26th to July 30th, 2025

5. Discussion and monitoring adaptations

5.1 Trophic status and sensitivity to climate change

Lake Taney's oxygen regime, points to a meso- to eutrophic deep-water system despite its oligotrophic surface appearance. Following Perga et al. (2023), under-ice oxygen decay in alpine lakes is primarily physically controlled, so persistent bottom hypoxia alone does not confirm elevated nutrient loading.

Three lines of evidence nonetheless point toward a nutrient-influenced system: the pronounced metalimnetic oxygen supersaturation and visible aquatic vegetation observed during the June 2026 transects ; the summer decay rate being nearly twice the winter rate, more consistent with temperature-dependent respiration of organic matter than sediment heat flux alone; and the depth-increasing conductivity/salinity signal indicating solute release from decomposing organic matter in the isolated hypolimnion. None is individually conclusive, confirming a nutrient signal requires sediment or productivity data the programme does not yet collect, but together they justify treating eutrophication risk as a first-order hypothesis for future monitoring, alongside the physical stratification dynamics already characterised.

Under climate warming, the mechanism is self-reinforcing: longer, stronger summer stratification (already suggested by the steeper 2025–2026 thermocline gradient relative to 2024–2025) combined with a shorter, less reliable ice season would

compress the window for full reoxygenation. The 2025-2026 record already illustrates this vulnerability, with spring mixing too brief and incomplete to meaningfully reoxygenate the deep basin. Continued shortening of ice duration could make incomplete spring turnover the norm, locking the hypolimnion into near-permanent low-oxygen conditions and raising the risk of internal phosphorus loading from anoxic sediments, independent of any change in external nutrient inputs from tourism or catchment land use.

5.2 Critical evaluation of the monitoring design

Strengths.

The dual-transect CTD sampling produced internally consistent vertical structure across both lake axes, confirming horizontal homogeneity and validating the single-mooring approach as representative of the wider basin. The sonar bathymetric survey, once drift-corrected, reproduced the known maximum depths and delivered a spatially resolved profile not previously available.

Limitations.

The single bottom-mounted DO sensor cannot distinguish genuine mixing from local sediment burial. A second MiniDOT at intermediate depth was planned to resolve this but could not be deployed after the spare unit failed at redeployment, the highest-priority instrumentation gap for the next campaign.

The pressure record's mid-January cutoff forced indirect, partly circular Chain B depth reconstruction thereafter, and compounds the 14 m S2-MiniDOT sensor gap responsible for the scattered thermocline-depth estimates in September-October 2025. The extended 30-minute pressure interval now implemented (below), and a planned intermediate temperature sensor would close both gaps. The pressure data show a water-level range of almost 6 m over the available record. This makes the pressure correction important for interpreting the real depth of the deep sensors and the reconstructed thermal structure.

Uncorrected lateral vessel drift compromised raw bathymetric acquisition, recovered only through post-hoc spatial smoothing. A more controlled acquisition method, a guideline between fixed shore points, or real-time GPS-guided steering, would reduce reliance on this correction and extend the method to Transect T3.

The interannual record spans only one complete ice-covered winter, sufficient to establish dimictic behaviour and a first decay-rate estimate, but not to distinguish interannual variability from a genuine trend in mixing efficiency, thermocline strength, or decay rate. At least two to three further annual cycles are needed.

5.3 Mooring modifications for the 2026-2027 deployment

The updated mooring deployed on 5 June 2026 retains the 2025-2026 configuration with three targeted modifications addressing the limitations above (Figure 27):

- An additional buoy was installed between the bottom weight and the MiniDOT logger to prevent the logger from settling into the sediment while keeping it submerged, directly addressing the sensor-burial ambiguity.
- The failed HOBO temperature logger at 11.4 m above the bottom was replaced with a Tinytag logger, recording every 20 minutes (versus 10 minutes, for the HOBO units) due to its lower memory capacity.
- The pressure logger's recording interval was extended from 15 to 30 minutes, to ensure the full deployment period through to the next retrieval, closing the mid-January gap that limited water-level correction this year.

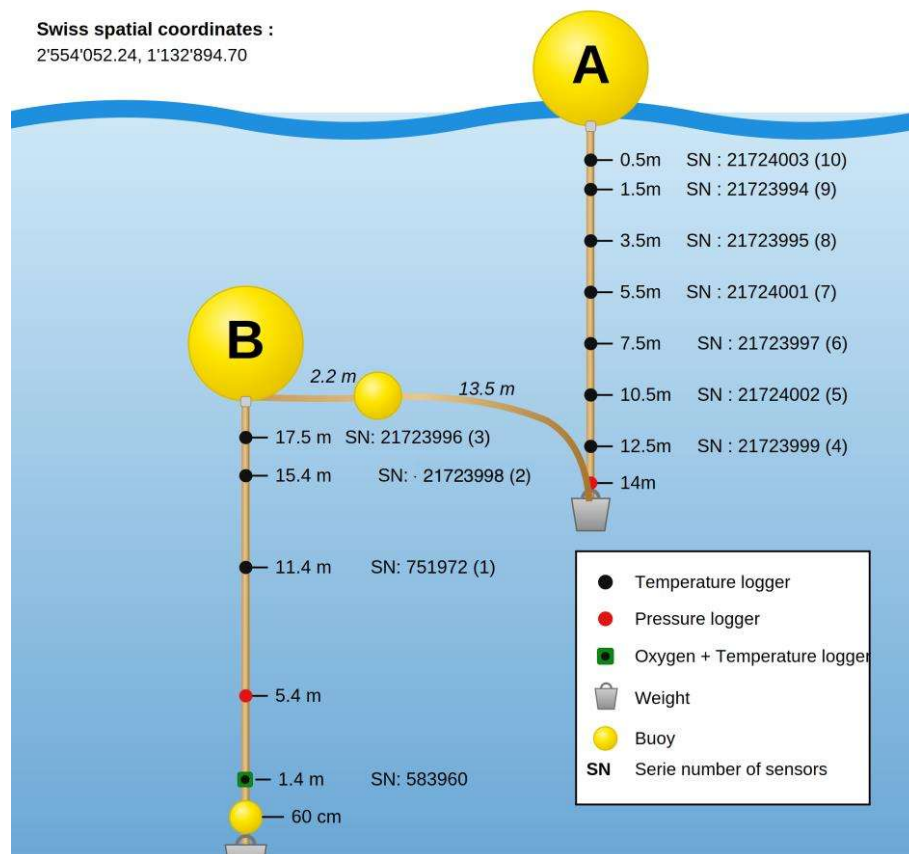


Figure 27: Updated mooring configuration diagram. Line A depths given from the surface; Line B depths given as height above the lake bottom. Coordinate (CH1903/LV03): 554069 / 132890

6. Conclusion and recommendations for future campaigns

6.1 Synthesis of key findings

The 2026 campaign confirms that Lake Taney functions as a dimictic mountain lake with tightly coupled, strongly seasonal thermal and oxygen regimes, but bottom waters under near-continuous ecological stress.

Thermally, the lake shows the seasonal structure typical of a dimictic regime, summer stratification, winter ice cover, and two mixing periods, but 2026 spring mixing was markedly shorter than autumn mixing suggesting that late ice-off can be rapidly overtaken by renewed stratification before the water column fully homogenises. This makes the deep-basin reoxygenation window directly sensitive to future changes in ice-cover duration, not merely to its timing.

Ecologically, legal oxygen thresholds were unmet for the large majority of the year and the under-ice decay rate ($k = 0.039 \text{ day}^{-1}$) places Lake Taney among the faster-depleting alpine lakes reported by Perga et al. (2023), offering a reproducible metric for tracking bottom-water conditions in future campaigns. The summer decay rate, nearly double the winter rate but less well constrained by an already-underway deployment, is not fully explained by the physical mechanisms known to dominate winter decay, resolving this gap should be a priority once productivity data are available.

Hydrologically, the close timing between the autumn recharge event and the sharp rise in bottom oxygen suggests autumn mixing is partly hydrologically, as well as thermally, driven. Because the pressure record ends in mid-January, this link cannot yet be tested for spring mixing, precisely where the mixing signal is most ambiguous and most in need of independent confirmation.

6.2 Recommendations for future monitoring

The following recommendations are grouped by priority and type, drawing together the individual points raised throughout the previous sections.

Instrumentation (highest priority)

- Deploy a working second MiniDOT sensor at an intermediate depth, pre-verified to avoid a repeat of the 2026 equipment failure, the main priority, enabling both resolution of the mixing-vs-burial ambiguity and depth-wise decay-rate comparison once a full annual cycle is available.
- Add an intermediate temperature sensor between S2 and the MiniDOT to close the 14 m blind zone affecting autumn thermocline estimates.
- Retain the extended 30-minute pressure-logger interval; review other loggers approaching memory limits before their scheduled retrieval.
- Add a retrieval loop or comparable fitting to the buoy to reduce field-handling time.

Methodology

- Anchor or guide the survey vessel along a fixed line during bathymetric/CTD transects rather than relying solely on post-hoc drift correction.
- Extend the sonar bathymetric survey to transect T3, now that the method is validated on T2, to build a yearly-updated bathymetric baseline.

New measurements

- Incorporate nutrient concentrations and biological indices (e.g. chlorophyll-a, macrophyte or macroinvertebrate surveys) to test the eutrophication hypothesis raised above.
- Having access to hydropower withdrawal data to separate artificial from natural water-level and volume variability
- Conduct a hypsometric survey to quantify heat storage and stratification-strength indicators from lake volume, better separating atmospheric heat-flux effects from snowmelt timing, precipitation and hydropower-driven throughflow.

Continuity

- Maintain consistent annual deployment and retrieval timing where possible, since the single-winter record currently available is insufficient to distinguish interannual variability from a longer-term trend in ice duration, mixing efficiency, or oxygen decay rates.

7. Contribution

Specific section detailing author's contribution to the common report (Who did what)

Author	Contribution
Bauermeister Lynn	Mooring deployment, CTD measurements along transect T2 including field recording, dissolved oxygen results interpretation and writing; supporting contribution to oxygen data processing notebook.
Bulotti Nicola	Mooring retrieval, transect 2 measurements, MiniDOT data processing, oxygen analysis (i.e. monthly/seasonal statistics, MiniDOT vs. HOBOT temperature comparison, Oxygen vs. Air Temperature/Wind) and notebook creation.
Criado Gabriela	Transect 2 recording measurements. Updated mooring deployment. Note taking, structuring and writing report (abstract, introduction, methods, results from previous campaigns, discussion, conclusion and recommendations, etc), Evaluation of gaps from previous report. Revision of the report, cutting, adjusting, etc.
Fringeli Coralie	Mooring retrieval, map of the transects and the mooring locations, Update of the mooring set up diagram and analysis of the temperature plots.
Grillon James	Mooring deployment, bathymetric transect recording, pressure data coding and analysis
Korkmaz Nihal	Data processing and analysis of dissolved oxygen (exponential bottom oxygen dynamics, temperature vs oxygen, monthly average etc.), interpretation of results. Updated mooring deployment.
Messerli Estelle	Weather data processing and analysis, Pressure data processing, README verification and update, code checking and corrections, GitHub submission, writing of corresponding report sections. Revision of the report, cutting, adjusting, etc.
Noël Quentin	Mooring retrieval, introduction/hydrological writing, map location adobe, bathymetric transect recording/analyses and plot of Sonar pro+2, transect T2 CTD measurement
Quillet T�rence	Mooring retrieval, temperature analysis notably stratification and ice cover onset/duration and atmospheric flux computation, redaction

Randrianasolo Iriana	Temperature data retrieval, computation and analysis, CTD data processing and analysis, Temperature sensor set-up, Mooring deployment. Writing for mooring set ups and improvements and CTD. Revision of the report, cutting, adjusting, etc.
Rottigni Nathan	Mooring retrieval, coding to adjust 2024 and 2025 campaign data temperature, temperature notebook, writing for graphs explanation, temperature methodology
Rouyard Cyprile	Mooring retrieval, measurements of the mooring, temperature analysis, summary of the important dates, analysis of the temperature plots.
Sanwald Giulia	Transect 2 measurements, pressure data coding and analysis, pressure section revision. Updated mooring deployment
Saravia Nicolle	Guevara Mooring retrieval, pressure data coding and analysis, metadata analysis CTD.

8. References

- Bickerstaffe, H., Bagnoud, N., Du Bois, R., Herold, P., Jacot-Descombes, R., Knight, J., Perez Pardo, A., Python, M., Pythoud, C., & Zaninetti, M. (2025). *Lake Taney Monitoring—Report 2025*.
- Canton du Valais. (2018). *Paysages, sites et monuments naturels—Fiches nature paysage—Vs.ch*. Fiches nature paysage. <https://www.vs.ch/web/fiches-nature-paysage/paysages-sites-et-monuments-naturels>
- Gabbud, J.-Y. (2021, June 22). Vouvry: *La protection de la région du lac de Taney renouvelée pour dix ans*. *Le Nouvelliste*. <https://www.lenouvelliste.ch/valais/chablais-valaisan/monthey-district/vouvry-commune/vouvry-la-protection-de-la-region-du-lac-de-taney-renouvelee-pour-dix-ans-1084605>
- geo.admin.ch. (2022). *Cartes de la Suisse—Confédération suisse—Map.geo.admin.ch*. https://map.geo.admin.ch/#/?lang=it¢er=2553640.6,1132913.42&z=9.653&topic=ech&layers=ch.swisstopo.zeitreihen@year=1864,f;ch.bfs.gebaeude_wohnungen_register,f;ch.bav.haltestellen-oev,f;ch.swisstopo.swisstlm3d-wanderwege,f;ch.vbs.schiessanzeigen,f;ch.astra.wanderland-sperrungen_umleitungen,f;ch.bafu.bundesinventare-bln;ch.bafu.vec25-seen;ch.swisstopo.swisstlm3d-gewaessernetz,f&bgLayer=ch.swisstopo.swissimage&featureInfo=default&catalogNo=ech
- Hayashi, M. (2004). Temperature-electrical conductivity relation of water for environmental monitoring and geophysical data inversion. *Environmental Monitoring and Assessment*, 96(1), 119-128. <https://doi.org/10.1023/B:EMAS.0000031719.83065.68>
- Jane, S. F., Mincer, J. L., Lau, M. P., Lewis, A. S. L., Stetler, J. T., & Rose, K. C. (2023). Longer duration of seasonal stratification contributes to widespread increases in lake hypoxia and anoxia. *Global Change Biology*, 29(4), 1009–1023. <https://doi.org/10.1111/gcb.16525>
- Kanamitsu, M., Ebisuzaki, W., Woollen, J., Yang, S.-K., Hnilo, J. J., Fiorino, M., & Potter, G. L. (2002). NCEP–DOE AMIP-II Reanalysis (R-2). *Bulletin of the American Meteorological Society*, 83(11), 1631–1644. <https://doi.org/10.1175/BAMS-83-11-1631>
- Kossmann, M., & Sturman, A. P. (2003). Pressure-driven channeling effects in bent valleys. *Journal of Applied Meteorology and Climatology*, 42(1), 151-158. [https://doi.org/10.1175/1520-0450\(2003\)042%3C0151:PDCEIB%3E2.0.CO;2](https://doi.org/10.1175/1520-0450(2003)042%3C0151:PDCEIB%3E2.0.CO;2)
- OFEV. (2022, November 28). *Inventaire fédéral des paysages, sites et monuments naturels (IFP)*. <https://www.bafu.admin.ch/fr/ifp>

OFEV (2022). *Vector25 lakes (ch.bafu.vec25-seen)*. Geo.admin.ch. <https://api3.geo.admin.ch/rest/services/ech/MapServer/ch.bafu.vec25-seen/41/extendedHtmlPopup?lang=it>

OFEV (2025, December 16). *Sites de reproduction de batraciens*. <https://www.bafu.admin.ch/fr/sites-reproduction-batraciens>

Perga, M., Minaudo, C., Doda, T., Arthaud, F., Beria, H., Chmiel, H. E., Escoffier, N., Lambert, T., Napolleoni, R., Obrador, B., Perolo, P., Rüegg, J., Ulloa, H., & Bouffard, D. (2023). Near-bed stratification controls bottom hypoxia in ice-covered alpine lakes. *Limnology and Oceanography*, 68(6), 1232–1246. <https://doi.org/10.1002/lno.12341>

Pierson, D. c., Weyhenmeyer, G. A., Arvola, L., Benson, B., Blenckner, T., Kratz, T., Livingstone, D. m., Markensten, H., Marzec, G., Pettersson, K., & Weathers, K. (2011). An automated method to monitor lake ice phenology. *Limnology and Oceanography: Methods*, 9(2), 74–83. <https://doi.org/10.4319/lom.2010.9.0074>

Pilla, R. M., Williamson, C. E., Adamovich, B. V., Adrian, R., Anneville, O., Chandra, S., Colom-Montero, W., Devlin, S. P., Dix, M. A., Dokulil, M. T., Gaiser, E. E., Girdner, S. F., Hambright, K. D., Hamilton, D. P., Havens, K., Hessen, D. O., Higgins, S. N., Huttula, T. H., Huuskonen, H., ... Zadereev, E. (2020). Deeper waters are changing less consistently than surface waters in a global analysis of 102 lakes. *Scientific Reports*, 10(1), 20514. <https://doi.org/10.1038/s41598-020-76873-x>

Pro Natura. (n.d.). *Taney | Pro Natura*. pronatura-vs.ch. Retrieved 4 June 2026, from <https://www.pronatura-vs.ch/fr/reserve-naturelle-de-taney-vouvry>

Rimmer, A., Aota, Y., Kumagai, M., & Eckert, W. (2005). Chemical stratification in thermally stratified lakes: A chloride mass balance model. *Limnology and Oceanography*, 50(1), 147-157. <https://doi.org/10.4319/lo.2005.50.1.0147>

Russo, N., Richter, N., Lattaud, J., & Dubois, N. (2025). *Seasonal Alkenone Production in a Mid-Latitude Alpine Lake*. 2025(1), 1–3. <https://doi.org/10.3997/2214-4609.202533154>

Swiss Confederation. (2023). *Waters Protection Ordinance (WPO) of 28 October 1998 (Status as of 1 February 2023)*. Fedlex. https://www.fedlex.admin.ch/eli/cc/1998/2863_2863_2863/en

Staeher, P. A., Bade, D., Van De Bogert, M. C., Koch, G. R., Williamson, C., Hanson, P., Cole, J. J., & Kratz, T. (2010). Lake metabolism and the diel oxygen technique :

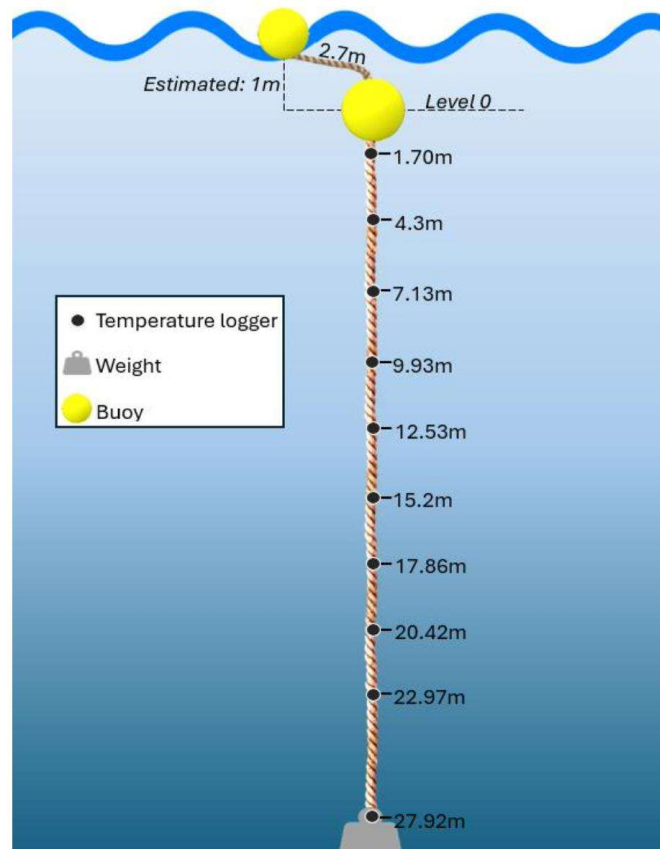
Staeher, P. A., Christensen, J. P. A., Batt, R. D., & Read, J. S. (2012). Ecosystem metabolism in a stratified lake. *Limnology and Oceanography*, 57(5), 1317-1330. <https://doi.org/10.4319/lo.2012.57.5.1317>

Vouvry. (1967). *Charte de Taney · Lac de Taney*. taney.ch. <https://taney.ch/charte-de-taney>

- Wells, M. G., & Troy, C. D. (2022). Surface mixed layers in lakes. In *Encyclopedia of Inland Waters* (p. 546-561). Elsevier. <https://doi.org/10.1016/B978-0-12-819166-8.00126-2>
- Wetzel, R. G. (2001). Fate of heat. In *Limnology* (p. 71-92). Elsevier. <https://doi.org/10.1016/B978-0-08-057439-4.50010-1>
- Williamson, C. E., Saros, J. E., Vincent, W. F., & Smol, J. P. (2009). Lakes and reservoirs as sentinels, integrators, and regulators of climate change. *Limnology and Oceanography*, 54(6part2), 2273–2282. https://doi.org/10.4319/lo.2009.54.6_part_2.2273
- Woolway, R. I., Kraemer, B. M., Lenters, J. D., Merchant, C. J., O'Reilly, C. M., & Sharma, S. (2020). Global lake responses to climate change. *Nature Reviews Earth & Environment*, 1(8), 388–403. <https://doi.org/10.1038/s43017-020-0067-5>
- Woolway, R. I., Maberly, S. C., Jones, I. D., & Feuchtmayr, H. (2014). A novel method for estimating the onset of thermal stratification in lakes from surface water measurements. *Water Resources Research*, 50(6), 5131–5140. <https://doi.org/10.1002/2013WR014975>

9. Annexes

Annex 1. Diagram of the old mooring from 2024 campaign, with a surface buoy connected to a submerged buoy (Level 0) above a bottom weight. The associated logger identifiers were, respectively: 2171001_0m at 1.70 m; 21724000_2.5m at 4.30 m; 21723998_5 at 7.13 m; 21724003_7.5 at 9.93 m; 21724002_10 at 12.53 m; 21723997_12.5m at 15.20 m; 21723999_15 at 17.86 m; 21723996_17.5 at 20.42 m; 21723995_20 at 22.97 m; and 21723994_25 at 27.92 m.



Annex 2: CTD coordinates (CH1903/LV03), sampling times, and maximum depth for Transect 2.

Location label	START_E	START_N	END_E	END_N	Time start [hours-min]	Time end [hours-min]	Depth [m]
T2.p1	554074	132746	554071	132745	11:30	11:31	7
T2.p2	554073	132757	534069	132753	11:33	11:37	11
T2.p3	554065	132770	554064	132756	11:38	11:42	20
T2.p4	554071	132787	554058	132781	11:44	11:48	30
T2.p5	554053	132794	554047	132792	11:50	11:52	32
T2.p6	554077	132853	554079	132853	11:54	11:57	33
T2.p7	554109	132915	554118	132929	12:03	12:05	33
T2.p8	554115	132942	554123	132956	12:06	12:08	28
T2.p9	554109	132963	554096	132971	12:12	12:15	22
T2.p10	554081	132975	554076	132970	12:15	12:17	13
T2.p11	554084	132983	554081	132982	12:18	12:20	8
T2.p12	554086	132588	554080	132989	12:21	12:22	6
T2.p13	554087	132988	554081	132989	12:24	12:26	5
T2.p14	554079	132993	554078	132988	12:27	12:29	3
T2.p15	554078	132992	554077	132991	12:31	12:33	4

Annex 3: CTD coordinates (CH1903/LV03), sampling times, and maximum depth for Transect 3.

Location label	START_E	START_N	END_E	END_N	Time start [hours-min]	Time end [hours-min]	Depth [m]
T3.p1	553430	132806	553440	132822	11:13	11:15	0.7
T3.p2	553532	132793	553543	132796	11:23	11:24	4.8
T3.p3	553674	132823	553673	132833	11:30	11:32	11.3
T3.p4	553804	132829	553798	132837	11:43	11:45	19.6
T3.p5	553894	132851	553888	132843	11:50	11:52	25.4
T3.p6	554028	132890	554010	132894	12:03	12:05	32.7
T3.p7	554052	132890	554043	132887	12:07	12:10	32.6
T3.p8	554095	132889	554079	132876	12:16	12:18	32.9
T3.p9	554161	132899	554159	132907	12:21	12:23	31.4
T3.p10	554220	132895	554218	132891	12:27	12:29	30.1
T3.p11	554253	132920	554245	132920	12:33	12:34	26
T3.p12	554283	132932	554292	132936	12:37	12:38	15.8
T3.p13	554292	132936	554325	132936	12:39	12:40	8.2
T3.p14	554332	132936	554369	132951	12:42	12:43	3.8